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NCERT

PARAKH

Guidelines for Setting Balanced Question Papers to Promote Equivalence in Assessment Practices of School Boards

Guidelines

for

SETTING BALANCED QUESTION PAPERS

to

PROMOTE EQUIVALENCE IN ASSESSMENT PRACTICES

of

SCHOOL BOARDS

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GUIDELINES FOR
SETTING BALANCED QUESTION PAPERS
TO PROMOTE EQUIVALENCE IN
ASSESSMENT PRACTICES OF SCHOOL BOARDS

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Shri Sanjay Kumar
Secretary

सत्यमेव जयते

Department of School
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Ministry of Education
Government of India

MESSAGE

Equitable and high-quality assessment is the cornerstone of a credible education system. The design of a question paper must reflect not only curricular intent but also a calibrated balance of cognitive demand, item diversity, and inclusivity. In the absence of a standardised framework, School Boards in the past had adopted approaches to question paper construction, resulting in inconsistencies that compromised on comparability and public trust.

The PARAKH Guidelines for Standardisation of Question Paper Templates (QPTs) offer a blueprint-driven approach to harmonise assessment design across Boards. By anchoring question papers in defined competencies, weightage norms, and validated item typologies, the guidelines promote structural equivalence while preserving contextual flexibility. This initiative will enable Boards to uphold rigor, transparency, and fairness in their assessment practices, thereby strengthening the integrity of school board examinations.

All School Boards are encouraged to adopt these guidelines in full measure, as a collective step towards a more coherent and outcome-aligned national assessment architecture.

Shri Anandrao V Patil
Additional Secretary

सत्यमेव जयते

Department of School
Education & Literacy,
Ministry of Education
Government of India

MESSAGE

The pursuit of equivalence in school assessment is not merely a technical exercise- it is a commitment to fairness, transparency, and national coherence in educational outcomes. As School Boards across India continue to evolve their pedagogical and assessment practices, the need for a harmonised framework for question paper design has become increasingly evident.

These Guidelines for Standardisation of Question Paper Templates (QPTs) provide a robust mechanism to support School Boards in constructing balanced, outcome-aligned question papers. By institutionalising blueprint-based design, the guidelines enable structural parity across diverse Boards and enhance the credibility of student performance data.

The Ministry recognises the diversity of educational contexts and encourages Boards to adopt these guidelines with due consideration to their operational realities. We remain committed to supporting this transition through capacity-building, technical assistance, and collaborative review mechanisms.

Together, let us advance towards a more equitable and accountable assessment ecosystem.

Foreword

The *Guidelines for Setting Balanced Question Papers to Promote Equivalence in Assessment Practices of School Boards* reflects a commitment to establishing fairness and equity in assessments across school boards in India. These guidelines aim to address longstanding disparities in question paper design by providing a structured framework that promotes consistency, inclusivity, and competency-based assessment practices.

Aligned with the vision of the National Education Policy 2020, the document emphasizes a shift from rote learning to methods that encourage critical thinking, problem-solving, and conceptual clarity. It highlights the importance of well-balanced question papers that are not only fair but also capable of evaluating students' abilities holistically. By standardizing the processes involved in question paper creation, these guidelines aim to ensure that assessments become meaningful tools for learning and growth.

This document is designed as a practical resource for educators, administrators, and policymakers, offering detailed steps and recommendations for setting balanced and effective question papers. It is hoped that the adoption of these guidelines will lead to a more robust and equitable education system, fostering transformative learning experiences for all students.

Prof. Dinesh Prasad Saklani
Director, NCERT

Preface

The *Guidelines for Setting Balanced Question Papers to Promote Equivalence in Assessment Practices of School Boards* is a comprehensive resource designed to address the pressing need for standardized and equitable assessment practices across school boards in India. Rooted in the principles of the National Education Policy (NEP) 2020, this document serves as a pivotal step in transitioning from traditional assessment models to competency-based systems that prioritize critical thinking, problem-solving, and conceptual clarity over rote learning.

This guideline document outlines a meticulously designed seven-step process for developing balanced and effective question papers. Each step is integral to creating assessments that are fair, reliable, and aligned with curricular goals. The process begins with the preparation of an assessment design, which focuses on defining the abilities, content areas, types of questions, and difficulty levels. This ensures that assessments are aligned with the curriculum and cater to diverse learning needs. The next step involves the development of an assessment blueprint, which acts as a detailed map, specifying the placement of questions according to abilities, content areas, and marks distribution. This step ensures coherence and consistency in the question paper.

The third step emphasizes question writing, which requires crafting questions that are clear, fair, and aligned with learning objectives. The guidelines stress the importance of clarity, unity, and economy in question design, alongside considerations for inclusivity and accessibility. This is followed by the assembly of the question paper, where questions are organized in a logical sequence to ensure a balance in content, difficulty, and format. The inclusion of clear instructions and section-wise structuring further enhances ease of administration.

Another critical aspect of the process is the development of a scoring key or marking scheme. To ensure objective and transparent assessment, the document provides detailed recommendations, including aligning marks with ability, recognizing valid alternative responses, and minimizing subjectivity in scoring. The sixth step, question paper analysis, involves a thorough review to identify any misalignments with the blueprint and design. Comprehensive checklists help ensure alignment with abilities, content, and question quality. Finally, the process concludes with moderation, which ensures that the question paper

adheres to prescribed standards of fairness, reliability, and validity by addressing biases, accessibility, and competency-based assessment principles.

One of the standout features of this document is its emphasis on fairness and inclusivity. The guidelines provide practical recommendations for eliminating cognitive and affective barriers, ensuring gender neutrality, and creating assessments that cater to diverse learning needs. The use of diverse question types, such as multiple-choice questions, short answers, and essays, allows for comprehensive assessment while accommodating different learning styles. Stimulus-based questions, marking schemes, and question paper analysis are also highlighted to enhance the overall quality of assessments.

This document serves as a valuable resource for educators, administrators, and policymakers, providing actionable insights and tools to transform assessment practices. By aligning with the broader goals of NEP 2020, this document paves the way for a future-ready education system that prioritizes fairness, equity, and competency-based learning.

I extend my sincere gratitude to all the school boards for their collaboration with PARAKH in finalizing this document.

Prof. Indrani Bhaduri
CEO and Head, PARAKH, NCERT

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1 Background

In July 2024, PARAKH released its first report on[1] the equivalence of school education boards. The report outlined a series of recommendations to bring greater parity among all recognized Indian educational boards across five broad categories: 1.) Administration, 2.) Curriculum, 3.) Assessment, 4.) Infrastructure, and 5.) Inclusiveness. These recommendations were informed by a detailed study of recent question papers set by various boards.

The section on Assessment examined Grade X and XII question papers from the previous academic year and identified several inconsistencies in assessment design, particularly in the following areas:

- **Inconsistent distribution of marks** relative to the cognitive demands of the content;
- **Uneven allocation of marks** according to the nature of the items; and
- **Lack of a systematic approach** to distributing marks across different difficulty levels.

Such disparities in assessment design present significant obstacles to achieving parity across boards and undermine equity in educational opportunities. A high-quality and robust assessment system is essential for ensuring educational equity. Recognizing this, NEP 2020 emphasized the need to establish equivalence across school education boards to raise assessment standards and foster fairness[2]. Restructuring assessment methodologies is also critical for reducing dependence on rote memorization and promoting critical thinking and the evaluation of core competencies.

Furthermore, establishing a greater level of equivalence among the question papers set by state boards will facilitate the ability of students to transfer between boards and schools will reduce disadvantages due to differing assessment systems and practices.

To address these challenges, PARAKH developed training sessions on designing balanced question papers and piloted the trainings with teachers and board members of the Himachal Pradesh Board of School Education (HPBoSE) during a two-day workshop at the HPBoSE campus in Dharamshala, Himachal Pradesh. Subsequently, a series of workshops on balanced question paper design were conducted at the PARAKH-NCERT campus in New Delhi. These sessions emphasized standardizing question paper development practices based on the principles of competency-based assessment.

[1] The Equivalence of School Education Boards Report
[2] National Education Policy 2020 [Section 4.41]

This guideline document will serve as a first step towards creating more balanced, consistent and higher quality assessments at the Board level. The recommendations are based on extensive interactions with workshop participants and the collective decade-long experience in education and test design, development, analysis, scaling, and reporting of the PARAKH team. Administrators, question writers, and educators should feel free to adapt the recommendations to their circumstances but should always keep in mind the goals of creating assessments that encourage student learning and growth for the challenges of the future.

2 Steps to Develop a Balanced Question Paper

In India, written exams are widely used and dominate the educational landscape. They are almost synonymous with evaluation, with all Educational Boards conducting annual and supplementary exams to assess students' achievement levels. Schools also hold their own terminal and annual exams. Given the importance of written assessments, it becomes essential to thoroughly examine the processes involved.

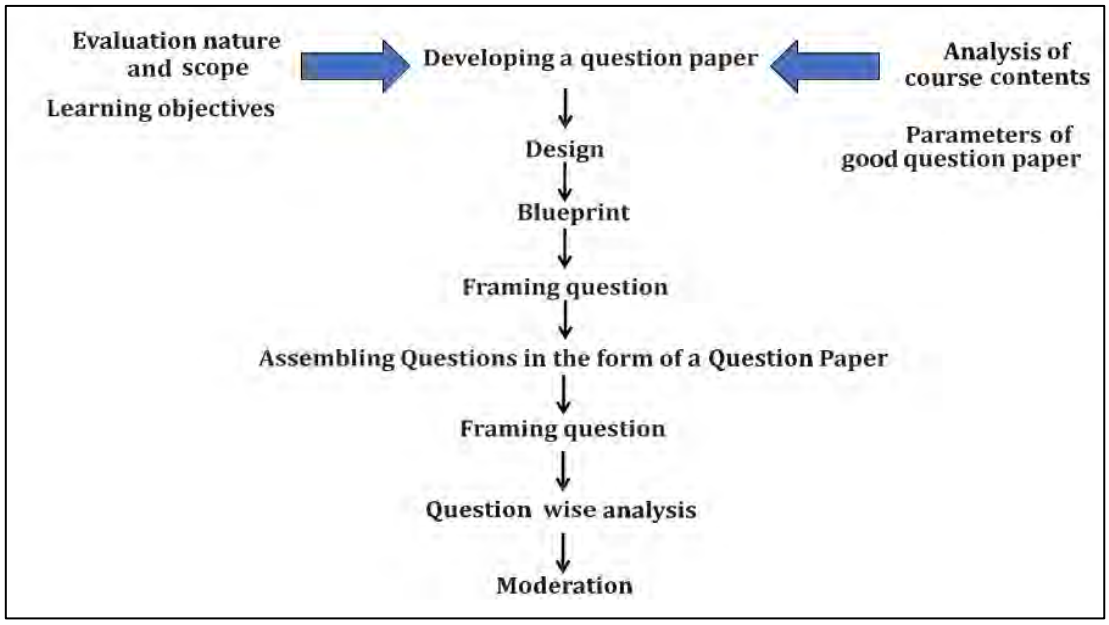
The question paper, a cornerstone of written exams, plays a crucial role in shaping learning outcomes. Its quality directly impacts student achievement and certification. Question papers also have strong influence on classroom teaching and pedagogy. Without an appropriate degree of standardization across time and student groups, variations in student performance across years and subjects will reflect both the quality of the question paper and genuine differences in student abilities, which limits the validity of claims based on question paper scores across time and student groups. For instance, a poorly designed or overly challenging paper may negatively affect overall achievement. Factors like content coverage, difficulty level, number of questions, types of questions, and the options provided must therefore be sufficiently balanced to ensure a fair assessment. Any imbalance in these areas can threaten the validity of claims based on assessment results. Creating a well-balanced question paper requires a systematic approach that incorporates design, blueprint creation, question formulation, marking schemes, and detailed analysis.

Once an examiner understands the characteristics of a good assessment, they can begin constructing it—whether it is a classroom unit test or a comprehensive question paper covering the entire syllabus. Regardless of the assessment type, key steps of assessment construction should include at minimum the following[3]:

1. Preparation of an Assessment Design
2. Preparation of an Assessment Blueprint.
3. Writing of Questions.
4. Assembling Questions in the form of a Question Paper.
5. Preparing Scoring Key/Marking Scheme.
6. Carrying out Question-wise Analysis.
7. Moderation of a Question Paper

[3] Please note, depending on the assessment purpose, the resources available, and the timelines for assessment development, inclusion of additional steps, such as small-scale pre-testing or large-scale piloting, may be advisable.

The chart below illustrates the process flow of a balanced question paper setting:



Note that different individuals or groups may be responsible for different steps in the process. Thus, clear documentation is essential.

2.1 Preparation of an Assessment Design (Step 1)

The first step in preparing a question paper is creating its design. A question paper is not merely a collection of random questions. It must be thoughtfully planned as an assessment, keeping in mind the objectives or abilities, the course content, and the forms of questions required to assess these. This involves assigning weightage to various objectives/abilities, content areas, and forms of questions while estimating the time required to answer each question, along with the scheme of sections and pattern of options. During the assessment design phase, the designers need to answer why they are planning to test, whom they are testing, what is being tested, what methods the test should include, what methods of quality control can be used during test design, and how much time, money, and other resources are available to design the test.

Test Design: Test design is the process of deciding how to meet the requirements established for a test within the unavoidable constraints that exist for the test.

The design serves as a framework of the question paper, much like an architectural map of a building. It determines how the completed question paper will appear and function. The design outlines essential components, such as weightage to different elements, to ensure alignment with the objectives of the test.

The figure below illustrates a design template for a question paper.

Format of Design (Subject other than languages)

Question Paper/Test

Subject	
Unit/Paper	
Class	
Time	
Marks	

Weightage to Objective/Cognitive Domains:

Objective	Awareness	Sensitivity	Creativity
Percentage of Marks			
Marks			

Weightage to Major Content Areas:

S. No.	Unit/Sub-Units	Marks
1.		
2.		
3.		
4.		
5.		
6.		
Total		

Weightage to form of Question:

Forms of Questions	E/LA	SA	VAS	O (MCQ)	Total
No. of Questions					
Marks allotted					
Estimated time					

		Beginner & Proficient	Advanced Level
Estimated difficulty level	Difficult	% Marks	% Marks
	Average	% Marks	% Marks
	Easy	% Marks	% Marks

Schemes of Section	
Pattern of Options	

Index of Abbreviation:

(E/LA: Essay/Long Answer; SA: Short Answer; VSA: Very Short Answer; O: Objective)

The design document should clearly state the following five major specifications:

1. Weightage to objectives/abilities
2. Weightage to major content areas
3. Weightage to types of questions
4. Weightage to different estimated difficulty levels
5. Scheme of sections and pattern of options

2.1.1 WEIGHTAGE TO OBJECTIVES/ABILITIES:

To ensure the validity of a test, it is crucial to analyze the objectives or abilities outlined in the course and determine which ones should be assessed and in what proportion. Marks are then assigned to each objective based on its significance, which is typically influenced by factors such as:

- The proportion of time spent covering that objective,
- The complexity and nature of the content in the course,
- The learning levels of the students

While assigning weightages to these objectives or abilities, question paper setters must align their decisions with the vision and guidelines of their respective boards. However, while setting these weightages, a question paper setter must be well informed about these abilities and how they are used in assessing the learning outcomes of the examinees. These abilities are *awareness (which includes Knowledge, understanding of the construct, understanding of the process), sensitivity (which includes Conflict Resolution, Open Mindedness, Collaboration) and creativity (which includes Generation of Innovative Ideas, Flexibility, Fluency, Exploration, Combining Idea & Concepts).* In the design, the weightages assigned to these abilities should be specified in both percentages and actual marks.

To ensure differentiated assessment aligned with students' readiness and aspirations, **a two-level examination structure** is proposed:

Beginner and Proficient Level: All students will undertake a standard examination assessing conceptual understanding, real-life application, and procedural fluency. The paper will include 25% Sensitivity, 50% Awareness, and 25% Creativity questions. Marks obtained will be reflected in the marks sheet.

Advanced Level: Students seeking to demonstrate higher-order thinking may opt for an additional 50-mark advanced examination. This will assess multi-concept integration,

strategic competence, and adaptive reasoning. Successful candidates will have their percentile and qualification mentioned in the certificate; unsuccessful attempts will not be recorded.

Below is an example to showcasing, how it could be filled for any subject .E.g. mathematics test of 100 marks of a secondary student:

Objective	Awareness	Sensitivity	Creativity	Total
Percentage of Marks (Beginner and Proficient)	50%	25%	25%	100%
Percentage of Marks (Advanced)	25%	25%	50%	100%
Marks (Beginner and Proficient)	50	25	25	100
Marks (Advanced)	12.5	12.5	25	50

Similarly, in language subjects as well, the focus would be on these three abilities only. For a 50-mark test, the weightages for these objectives can be distributed accordingly in percentages. Below is an example of a possible distribution:

Objective	Awareness	Sensitivity	Creativity	Total
Percentage of Marks (Beginner and Proficient)	50%	25%	25%	100%
Percentage of Marks (Advanced)	25%	25%	50%	100%
Marks (Beginner and Proficient)	25	12.5	12.5	50
Marks (Advanced)	6.25 (approx.)	6.25 (approx.)	12.5	25

2.1.2 WEIGHTAGE TO MAJOR CONTENT AREAS

It is equally important to analyze the syllabus and assign weightages to different areas of content to ensure the test's validity. The weightage for each content area is determined by factors such as the amount of instructional time dedicated to it during the academic year.

To illustrate, consider a scenario where an academic year includes 100 instructional periods to complete 10 units or sub-units of a subject, which will have a final assessment of 80 marks Instead of distributing marks equally (e.g., 8 marks per unit), the weightages should reflect the proportion of time required to cover each unit/sub-unit out of total available time for course completion in an academic year. This ensures a fair representation of the content's relative importance. The weightage to unit/sub-unit is calculated using the following relation:

This implies that if unit/sub-unit 1 takes 5 periods to complete while unit 2 takes 15 periods to complete. Then,

Weightage of unit 1 should be, $= \frac{5}{100} \times 100 = 5$ Marks

Weightage percentage to the unit 1 $= \frac{4}{80} \times 100\% = 5\%$

And weightage of Unit 2 $= \frac{15}{100} \times 100 = 15$ Marks

Weightage percentage to the unit 2 $= \frac{12}{80} \times 100\% = 15\%$

In addition, the following factors should also be considered when deciding the weightage for different content areas:

- Units/sub-units that build foundational concepts for senior classes or competitive exams should be given greater weightage,
- Units or sub-units with more complex learning objectives could receive higher weightage if desirable,
- The density of content elements within each unit or sub-unit should also influence the assigned weightage.

2.1.3 WEIGHTAGE TO TYPES OF QUESTIONS

After analyzing the objectives and content, it is essential to determine the most suitable ways to assess them. Different objectives and content areas are better tested using specific forms of questions. Therefore, a variety of question types should be included in the test, along with a clear allocation of marks and estimated time for each type. This process enhances the reliability of the test.

A question paper setter may consider the following points, while deciding the weightage to different forms of questions:

- Within a particular group of the same type of question, the time and marks distribution can vary. For example, in a question paper, a MCQ of awareness domain may carry 1 mark for instance, whereas a MCQ targeting higher-order cognitive skills (such as analyze or evaluate) could carry 2 or 3 marks.
- Similarly, questions of differing estimated difficulty levels within the same type may have varying marks. For example, a question judged to be easy may carry fewer marks than a question judged to be difficult even if they are of the same type.

- The time allocated to answer a question depends on its cognitive demand and estimated difficulty level.

2.2 Weightage to Different Estimated Difficulty Levels

When planning a question paper, it is important to include questions of varying difficulty levels to address the needs of all students and to differentiate between high and low achievers. The difficulty level of a question should align with the abilities and achievement levels of the target students.

The question paper setter may estimate the difficulty level of a question based on:

- The complexity of mental processes involved.
- The specific content area being assessed.
- The time required to answer it.
- Observed difficulty of similar questions on previous exams

Because of the difficulty of accurately predicting difficulty levels of test questions, preliminary estimates of difficulty are best used as a rough sorting device. Question writers and reviewers are advised to adopt a simple three-level scheme for estimated difficulty (Easy, Medium, and Hard) and use generous ranges rather than precise numbers for each level. The true difficulty level is determined only after the question is administered and analyzed.

2.2.1 SCHEME OF SECTIONS

The design of a question paper may also indicate the scheme of sections for the paper. For example, a question paper may consist of both Selected response questions and Constructed-response questions. Such a test may have two sections, one consisting of Selection-type questions like MCQs, Alternative choice questions (True-False), and Matching type questions. While the other consists of constructed-response questions like essay or long answer type, short answer and very short answer type questions.

Similarly, for non-language subjects, sections might be organized by content areas, such as Physics, Chemistry, and Biology for Science, or History, Geography, and Civics for Social Studies. In language subjects, sections could be based on skill areas, such as reading comprehension, writing tasks, and grammar.

2.2.2 PATTERN OF OPTIONS (QUESTION CHOICE)

The design should indicate the pattern of options i.e. whether there are any options between the questions in the question paper, if so, how many and what their nature is. From the academic point of view, there is no justification for providing options except to make the test

actually or apparently simpler for the examinees. The best comparison of achievement of the examinees can be made only if all of them are put to the same test without any choices. Thus, test-takers should not be allowed to choose a subset of questions choices to attempt. For example, if a question paper allows students to attempt any 25 out of 40 questions, some objectives or content areas may go untested or receive limited responses, thus compromising the test's reliability.

However, if there is a mandate to provide some type of question choice, it is best to provide limited **internal choice** within selected questions (typically constructed response questions). For instance, a constructed response question on a history question paper might give the student a small choice of significant events of the same type and time period to write about.

When providing such a choice, the alternatives must:

- Be comparable in terms of the objectives being tested.
- Maintain consistency in question format and difficulty level.
- Cover similar content areas as far as possible.

The statistical performance of questions featuring internal choice should be monitored. Any significant score discrepancies or difficulties scoring these questions should be caused to reconsider the policy of offering choice.

2.2.3 FORMAT OF DESIGN

The question paper design is typically created by the Examining Board in consultation with subject experts, teachers, and examiners. Decisions on weightage to content units are made by a committee appointed by the Board administration. Once finalized, the completed design template is provided to paper setters for implementation.

Design templates for language and non-language subjects are provided below for reference:

Question Paper/Test

Class: _____

Subject: _____ Unit/Paper: _____

Time: _____ Marks: _____

Weightage to Objectives

Objective	Awareness	Sensitivity	Creativity	Total
Percentage of Marks (Basic and Proficient)	50%	25%	25%	
Percentage of Marks (Advanced)	25%	25%	50%	
Marks				

Weightage to Major Content Areas:

S.No.	Unit/Sub-Units	Marks
1.		
2.		
3.		
4.		
5.		
6.		
Total		

Weightage to form of question:

Forms of Questions	E/LA	SA	VSA	O (MCQ)	Total
No. Of Questions					
Marks allotted					
Estimated time					

		Beginner and Proficient	Advanced
Estimated difficulty level:	Difficult	25%	50%
	Average	50%	25%
	Easy	25%	50%

Schemes of Sections	
Pattern of Options	

Index of Abbreviations:
(E/LA: Essay/Long Answer; SA: Short Answer; VSA: Very Short Answer; O: Objective)

2.3 Preparation of an Assessment Blueprint (Step 2)

The preparation of a blueprint is a crucial step after finalizing the test design. A blueprint serves as a detailed plan that clearly maps out the placement of each question based on its ability/objective, content area, form, sequence number, and allocated marks. A well-constructed blueprint helps bring the test design to life in practical terms. It goes beyond merely assigning weightage to objectives, content areas, and question types, as it specifies the exact relationship between these elements. This ensures that the test maker has a clear understanding of which question assesses which ability/objective, from which content unit, through which form of question, weightage it carries, and at what position it will appear in a question paper.

Creating a blueprint involves technical skill and subject-specific insight. Paper setters must have a clear understanding of the content hierarchy, the learning outcomes associated with each unit, and how these align with assessment objectives. This clarity enables them to decide on the most suitable types of questions for each unit and the abilities/objectives these questions will assess.

A blueprint represents the overall structure of the question paper, including:

- Distribution of marks across abilities/assessment objectives,
- Distribution of marks across content units,
- Types of questions and their weightage.
- A summary section.

A thoughtfully prepared blueprint not only ensures alignment with the prescribed weightages of test design but also serves as a comprehensive guide for creating a balanced and effective question paper.

2.3.1 Functional Definitions

Abilities		Functional Definition
AWARENESS	**Knowledge**	The ability to recall, recognize, identify, realize and cognizance of learned facts, concepts, principles, and procedures relevant to a construct.
	Understanding of the Construct	The ability to describe, analyse, synthesise, interpret, concept map, and make sense of the key ideas, principles, or features that define a particular concept or domain, demonstrating comprehension beyond simple recall.
	Understanding of the Process	The ability to narrate, enumerate, chart, sequence, and apply the steps, methods, and procedures necessary to accomplish a task, solve a problem, or conduct an inquiry within a domain.
SENSITIVITY	**Conflict Resolution**	The ability to acknowledge and address conflicting situations within self, among individuals or groups by identifying the type and source of conflict, applying appropriate negotiation strategies, and encouraging constructive outcomes. This includes accepting situations when constructive conflicts occurs and using integrative approaches to resolve different perspectives and achieve shared goals.
	Open Mindedness	The ability to consider new ideas, perspectives, and evidence with impartiality and respect, and to revise one's own disposition, beliefs or assumptions when warranted. Open-mindedness involves a readiness to critically examine one's own views, accept differences, and seek understanding in the pursuit of truth and knowledge.
	Collaboration	The ability to work effectively and respectfully with others to achieve a common goal, demonstrating flexibility, shared responsibility, and valuing diverse contributions. Collaboration includes identifying situations that require group problem-solving and decision-making, exercising willingness to accept others view point, and applying strategies that support productive teamwork.
CREATIVITY	**Generate Innovative Ideas**	The ability to go beyond routine problem-solving by encouraging students to think divergently, generate original ideas, and apply knowledge in novel ways across different contexts.
	Flexibility	The ability to generate a variety of ideas, shift between different approaches, and use multiple strategies when addressing tasks or solving problems across contexts
	Fluency	The ability to rapidly generate a large number of relevant ideas, solutions, or responses to a given prompt or problem, demonstrating ease and flexibility in thinking across various contexts.
	Exploration	The ability to examine situations from multiple perspectives, actively question and challenge assumptions, and seek deeper understanding or alternative explanations across contexts.
	Combining Idea & Concepts	The ability to integrate and combine ideas or concepts into coherent explanations or solutions across various contexts or disciplines.

2.3.2 STEPS TO DEVELOP A BLUEPRINT

The following is a recommended templet for a question-paper blueprint:

FORMAT OF BLUEPRINT

Subject: _____ **Class:** _____

Unit/Paper: _____ **Max Marks:** _____ **Time:** _____

Objective	Awareness				Sensitivity				Creativity					Total Row Wise	
Form of Questions	Knowledge	Understanding of the construct	Understanding of the process	Conflict Resolution	Open Mindedness	Collaboration	Generate Innovative Ideas	Flexibility	Fluency	Exploration	Combining Idea & Concepts				
Content Unit/ Sub Unit															
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	LA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	SA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	VSA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	O			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	LA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	SA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	VSA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	O			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	LA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	SA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	VSA			
	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	d e f g	O			
Sub Total														Marks (QS)	
Total Column Wise	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)	Marks Total (b)			(a)	
Total	Marks Total														Marks Total

Notes: - Figures within brackets to indicate the number of questions and figures outside the brackets to indicate marks.
- Denotes those marks have been combined to form one question.

SUMMARY SHEET

Summary: **Long Answer (LA)** **No.** _____ **Marks** _____ **Pattern of Options** **Grammar (One-word QS) and Writing Section**

Short Answer (SA) **No.** _____ **Marks** _____

Very Short Answer (VSA) **No.** _____ **Marks** _____ **Scheme of Sections** **Content Area**

Objective (O) **No.** _____ **Marks** _____

S. No.	Difficult Level	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

S. No.	Abilities	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

The development of a blueprint is a sequential process that makes the job a bit organised and easy for a question paper setter. The small letters a, b, c, d, e, f, and g in the given template are not part of the blueprint itself. They are steps to guide question paper setters. The details of these steps are the following:

I. Begin by filling in the top section of the proforma:
   - Mention the subject, class, maximum marks, time, and name of the examination for which the blueprint is being prepared.

II. **Step (a):** Write the maximum marks for the exam in the bottom-right corner of the proforma.

III. **Step (b):** Record the weightages for abilities or objectives, as decided in the test design.

IV. **Step (c):** Enter the weightages assigned to various content units/sub-units in the rightmost column.

V. **Steps (d) to (g):**
- Use these columns to input details for different types of questions (Essay/Long Answer, Short Answer, Very Short Answer, and Objective).
- Fill the cells using the format: Question Sequence (Marks). For example, "Q1 (5)" means question 1 positioned in the paper carries 5 marks.
- Leave blank cells marked with a <-> to indicate no questions.
- Use asterisks (*) and dotted lines to show marks split between two objectives.
- Shade cells where internal choices are provided between the questions.

VI. **Verify the Question Paper Design**
- Count the number of questions for each type (Essay, Short Answer, etc.) and ensure it matches the design.
- Check row and column subtotals to confirm alignment with weightages for abilities/objectives and content units.
- Adjust the blueprint if any discrepancies arise.

VII. **Finalize the Blueprint**
- Fill in the summary section, which includes the question types, section scheme, and pattern of question choice (if applicable).

2.3.3 STEPS FOR QUALITY ANALYSIS OF A BLUEPRINT

After completing the blueprint, review the following:

I. Does it align with the weightages for objectives, content units, and question types as outlined in the design?
II. Are the predisposing conditions, the pattern of question choice, and the number and form of questions followed correctly?
III. Are questions evenly distributed across content areas and abilities?
IV. Are internal choices balanced in terms of content, difficulty, and time requirements?

A blueprint translates the test design into an actionable plan. It supports validity and reliability by guiding question setters to create various types of questions that align with the test's goals. While a good blueprint is crucial, the quality of the final question paper depends on how well the questions adhere to the blueprint.

The blueprint templates for all subjects are attached below for reference:

SUBJECT WISE FORMAT OF BLUEPRINT

FORMAT OF BLUEPRINT
Language R1 (Secondary Stage)

Subject: _____ **Class:** _____

Unit/Paper: _____ **Max Marks:** _____ **Time:** _____

Objective	Awareness												Sensitivity												Creativity																				Total Row Wise
Form of Questions	Knowledge				Understanding of the construct				Understanding of the process				Conflict Resolution				Open Mindedness				Collaboration				Generate Innovative Ideas				Flexibility				Fluency				Exploration				Combining Idea & Concepts				
Content Unit/ Sub Unit	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	LA	SA	VSA	O	
Sub Total																																													Marks (QS)
Total Column Wise	Marks Total				Marks Total				Marks Total				Marks Total				Marks Total				Marks Total				Marks Total				Marks Total				Marks Total				Marks Total							
Total	Marks Total												Marks Total												Marks Total																				

Notes: - Figures within brackets to indicate the number of questions and figures outside the brackets to indicate marks.
- Denotes those marks have been combined to form one question.

SUMMARY SHEET FOR LANGUAGE (R1)

Summary: **Long Answer (LA)** **No.** _____ **Marks** _____ **Pattern of Options** **Grammar (One-word QS) and Writing Section**

Short Answer (SA) **No.** _____ **Marks** _____

Very Short Answer (VSA) **No.** _____ **Marks** _____ **Scheme of Sections** **Content Area**

Objective (O) **No.** _____ **Marks** _____

S. No.	Difficult Level	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

S. No.	Abilities	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

COMPETENCIES COVERED: LANGUAGE (R1)

S. No.	Competency Code	Competency Details	Question Number	No. of items
1	**C-1.1**	Uses language appropriate to social context, expresses agreement and disagreement with reasons, and arrives at conclusions through discussion and debate		
2	**C-1.2**	Writes in different styles (narrative, descriptive, expository, and persuasive) from their own, as well as others experiences		
3	**C-1.3**	Writes for real-life situations (invitations, speeches, condolence messages, notices, creative slogans, advertisements, etc.) and for school newsletter/magazine/journal		
4	**C-1.4**	Scripts to inform and communicate ideas effectively with the use of technology		
5	**C-2.1**	Describes characteristics of works of literature from different time periods, such as, early, medieval, and contemporary		
6	**C-2.2**	Analyses a literary text by close reading, critiquing form and style, and interpreting possible meanings		
7	**C-2.3**	Composes literary texts by using appropriate literary devices		
8	**C-3.1**	Analyses and evaluates the different audio and written material		
9	**C-3.2**	Argues with proper rationale by carefully evaluating premises		
10	**C-4.1**	Recognises the multilingual nature of Indian society and richness of its literary works through reading texts and watching content of different genres		
11	**C-4.2**	Appreciates the richness of culture and heritage in the different works of regional languages and with each other their connections		
12	**C-4.3**	Shows an understanding of the role of language in the formation of our identities and culture		
13	**C-4.4**	Demonstrates a basic knowledge of the commonalities among some of the major Indian languages, such as, their common phonetic and scientifically arranged alphabets and scripts, common grammatical structures, origins and sources of vocabularies from Sanskrit and other classical languages, as well as their rich interinfluences and differences		
14	**C-4.5**	Demonstrates a basic knowledge of languages spoken in different geographical areas, a sense of the nature and structure of tribal languages, and becomes familiar with a few useful words phrases, and works of literature from a few Indian languages from across the country		

SUMMARY SHEET FOR LANGUAGE (R2)

S. No.	Abilities	Standard for Basic & Proficient Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

Nature of Questions	Standard for Basic, Proficient & Advance	No. of Items	Marks
Long Answer (LA)	35%		
Short Answer (SA)	30%		
Very Short Answer (VSA)	15%		
Objective (O)	20%		

S. No.	Difficult Level	Standard for Basic & Proficient Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

COMPETENCIES COVERED: LANGUAGE (R2)

S. No.	Competency Code	Competency Details	Question Number	No. of items
1	**C-1.1**	Uses language appropriate to social context, expresses agreement and disagreement with reasons, and arrives at conclusions through discussion and debate		
2	**C-1.2**	Writes in different styles (narrative, descriptive, expository, and persuasive) from their own as well as others experiences		
3	**C-1.3**	Writes for real-life situations (invitations, speeches, condolence messages, notices, creative slogans, advertisements, etc.) and for school newsletter/ magazine/journal.		
4	**C-1.4**	Scripts to inform and communicate ideas effectively with the use of technology		
5	**C-2.1**	Analyses and evaluates different audio and written material		
6	**C-2.2**	Argues with proper rationale by carefully evaluating premises		
7	**C-3.1**	Describes characteristics of works of literature from different time periods, such as, early, medieval, and contemporary)		
8	**C-3.2**	Analyses a literary text by close reading, critiquing form and style, and interpreting possible meanings		
9	**C-3.3**	Composes literary texts using appropriate literary devices		

SUMMARY SHEET FOR LANGUAGE (R3)

Summary: **Long Answer (LA)** **No.** _____ **Marks** _____ **Pattern of Options** **Grammar (One-word QS) and Writing Section**

Short Answer (SA) **No.** _____ **Marks** _____

Very Short Answer (VSA) **No.** _____ **Marks** _____ **Scheme of Sections** **Content Area**

Objective (O) **No.** _____ **Marks** _____

S. No.	Difficult Level	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

S. No.	Abilities	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

COMPETENCIES COVERED: LANGUAGE (R3)

S. No.	Competency Code	Competency Details	Question Number	No. of items
1	**C-1.1**	Identifies main points, summarises after a careful reading of the text, and responds coherently		
2	**C-1.2**	Uses strategies to organise ideas and information to write for an intended purpose and audience		
3	**C-2.1**	Listens critically and reads different news articles, reports and editorials to express opinions		
4	**C-2.2**	Asks a variety of questions on social experiences using appropriate language (open-ended/closedended, and formal/informal, questions relevant to context, with sensitivity)		
5	**C-2.3**	Shares ideas and critiques on the various aspects of their social and cultural surroundings in oral and written form		
6	**C-2.4**	Writes different kinds of letters and essays in an appropriate language for different audiences		
7	**C-3.1**	Identifies and appreciates different forms of literature, such as, samples of prose, poetry and drama (early to contemporary)		
8	**C-4.1**	Interprets, understands, and applies basic linguistic aspects (rules), such as, sentence structure, punctuation, tense, gender, and parts of speech		
9	**C-5.1**	Understands the phonetics and script of the language, the number of vowels and consonants, and how they interact and are used		
10	**C-5.2**	Engages in the use of puns, rhymes, alliteration and other wordplays in the language to make speech and writing more interesting and enjoyable		
11	**C-5.3**	Becomes familiar with some of the major word games in the language (e.g., palindromes, spoonerisms, sentences without given letters or sounds, riddles, jokes, antakshari, anagrams, crosswords, etc.)		

SUMMARY SHEET FOR SOCIAL SCIENCE

S. No.	Abilities	Standard for Basic & Proficient Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

Nature of Questions	Standard for Basic, Proficient & Advance	No. of Items	Marks
Long Answer (LA)	30%		
Short Answer (SA)	30%		
Very Short Answer (VSA)	20%		
Objective (O)	20%		

S. No.	Difficult Level	Standard for Basic & Proficient Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

COMPETENCIES COVERED: SOCIAL SCIENCE

S. No.	Competency Code	Competency Details	Question Number	No. of items
1	**C-1.1**	Explains historical events and processes using different types of sources, with specific examples from Indian history		
2	**C-1.2**	Explains and analyses the chronology of human life on the Indian subcontinent, from prehistory to its civilisational beginnings and beyond, and its relations with other civilisations over time, such as, those in Mesopotamia, Greece, Central Asia, China, South-east Asia, Arabia, and Eastern Africa.		
3	**C-1.3**	Traces aspects of continuity and change in different phases of history across the Indian subcontinent (including cultural trends, social and religious trends and reforms, and economic and political transformations)		
4	**C-1.4**	Explains the growth of new indigenous ideas across India including in Mathematics, Philosophy, Science and Technology, Medicine, Architecture, Agriculture, Literature and Art, and Social Science (such as, zero and the Indian number system, Ahimsa, the six systems of Indian philosophy, Ayurveda, yoga, the 22 Shrutis of Indian music, horticulture, use of herbs and spices, etymology, metres, and grammar) and how they affected the course of Indian and world history		
5	**C-2.1**	Explains historical events and processes with different types of sources, with specific examples from world history		
6	**C-2.2**	Explains and analyses the chronology of human life from its beginnings to nomadism to settled life and other phases of human civilisation		
7	**C-2.3**	Traces aspects of continuity and change in different phases of world history (including cultural trends, social and religious reforms, and economic and political transformations)		
8	**C-2.4**	Explains the growth of new ideas and practices across the world (including humanism, mercantilism, industrialisation, scientific developments and explorations, imperialism, colonialism, the rise of new nation-states across the world, and various technologies including the most current) and how they affected the course of world history		
9	**C-2.5**	Recognises the various practices that arose, such as, those in C-2.4, and came to be condemned later on (such as, racism, slavery, colonial invasions, conquests, and plunder, genocides, exclusion of women from democratic and other institutions), all of which have also impacted the course of world history and have left unhealed wounds		

S. No.	Competency Code	Competency Details	Question Number	No. of items
10	**C-3.1**	Analyses the meaning of nation and how the concept evolved over time across the world and in the specific context of India, including its roots in the rich civilisational history of the Indian subcontinent		
11	**C-3.2**	Identifies and analyses important phases of the Indian national freedom struggle against British colonial rule, with special reference to the movement led by Mahatma Gandhi and other important figures as well as those that led to independence, and understands the specific Indian concepts, values and methods (such as, Swaraj, Swadeshi, passive resistance, fight for Dharma, self-sacrifice, Ahimsa) that played a part in achieving Independence		
12	**C-4.1**	Locates physiographic regions of India and the climatic zones of the world on a globe or map		
13	**C-4.2**	Explains important geographical concepts, characteristics of key landforms, their origin, and other physical factors of a region		
14	**C-4.3**	Draws inter-linkages between various components of the physical environment, such as, climate and relief, climate and vegetation, vegetation and wildlife		
15	**C-4.4**	Analyses and evaluates the inter-relationship between the natural environment and human beings and their cultures across regions and, in the case of India, the special environmental ethos that resulted in practices of nature conservation		
16	**C-4.5**	Critically evaluates the impact of human interventions on the environment, including climate change, pollution, shortages of natural resources (particularly water), and loss of biodiversity; identifies practices that have led to these environmental crises and the measures that must be taken to reverse them		
17	**C-4.6**	Develops sensitivity towards the judicious use of natural resources (by individuals, societies, and nations) and suggests measures for their conservation		
18	**C-5.1**	Understands that the Indian Constitution draws from the great cultural heritage and common aspirations of the Indian nation, and recalls India's early experiments with democracy (assemblies in Mahajanapadas, kingdoms and empires at several levels of the society, guilds, Sanghas and Ganas, village councils and committees, Uthiramerur inscriptions)		
19	**C-5.2**	Appreciates fundamental Constitutional values and identifies their significance for the prosperity of the Indian nation		

S. No.	Competency Code	Competency Details	Question Number	No. of items
20	**C-5.3**	Explains that fundamental rights are the most basic human rights, and they flourish when people also perform their fundamental duties		
21	**C-5.4**	Analyses the basic features of a democracy and democratic government and its history in India and across the world and compares this form of government with other forms of government		
22	**C-5.5**	Analyses the critical role of non-state and nonmarket participants in the functioning of a democratic government and society, such as, the media, civil society, socio-religious institutions, and community institutions		
23	**C-6.1**	Understands how the Indian ethos and the cultural integration across India did not attempt uniformity, but respected and promoted a rich diversity in Indian society, and how this harmonisation and unity in diversity, with a historical respect for all cultures, women have counted among India's great strengths by promoting peaceful coexistence		
24	**C-6.2**	Understands that, despite C-6.1, forms of inequality, injustice, and discrimination have occurred in different sections of society at different times (due to internal, as well as, outside forces, such as, colonisation), leading to political, social, and cultural efforts, struggles, movements, and mechanisms at various levels towards equity, inclusion, justice, and harmony, with varying outcomes and degrees of success		
25	**C-6.3**	Analyses aspects of differential treatment or discrimination that may exist in Indian society, based on, e.g., socio-cultural background, region, language spoken, and what individuals and societies can do to eradicate such differential treatment		
26	**C-6.4**	Understands that a progressive society and nation, such as, India is one that recognises not only its civilisational strengths but also its socio-economic, cultural, and political challenges and continuously makes efforts to address those challenges to become ever more prosperous, inclusive, just, and harmonious		
27	**C-7.1**	Defines key features of the economy, such as, production, distribution, demand, supply, trade, and commerce, and factors that influence these aspects (including technology)		
28	**C-7.2**	Evaluates the importance of the three sectors of production (primary, secondary, and tertiary) in any country's economy, especially India		
29	**C-7.3**	Distinguishes between 'unorganised' and 'organised' sectors of the economy and their role in production for the local market in small, medium, and large-scale production centres (industries), and recognises the special importance of the so-called 'unorganised' sector in Indian economy and its connections with the self-organising features of Indian society		

S. No.	Competency Code	Competency Details	Question Number	No. of items
30	**C-7.4**	Traces the beginning and importance of large-scale trade and commerce (including e-commerce) between one country and another, the key items of trade in the beginning, and the changes from time to time		
31	**C-8.1**	Gathers, comprehends, and analyses data related to income, capital, poverty, and employment in one's locality, region and at the national level		
32	**C-8.2**	Understands and analyses the concepts and practice of the range of economic systems from free market to entirely state controlled markets		
33	**C-8.3**	Understands these features in the context of ancient India, with its thriving trade, both internal and external, and its well-established trade practices and networks, business conventions, and diverse industries, all of which made India one of the world's leading economies up to the colonial period		
34	**C-8.4**	Describes India's recent path towards again becoming one of the three largest economies of the world, and how individuals can contribute to this economic progress		
35	**C-8.5**	Appreciates the connections between economic development and the environment, and the broader indicators of societal well-being beyond GDP growth and income		
36	**C-9.1**	Knows and explains the significant contributions of India to all matters (concepts, explanations, methods) studied within the curriculum, in an integrated manner		

SUMMARY SHEET FOR SCIENCE

S. No.	Abilities	Standard for Basic & Proficient Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

Nature of Questions	Standard for Basic, Proficient & Advance	No. of Items	Marks
Long Answer (LA)	25%		
Short Answer (SA)	30%		
Very Short Answer (VSA)	20%		
Objective (O)	25%		

S. No.	Difficult Level	Standard for Basic & Proficient Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

COMPETENCIES COVERED: SCIENCE

S. No.	Competency Code	Competency Details	Question Number	No. of items
1	**C-1.1**	Describes classification of elements in the Periodic Table, and explains how compounds (including carbon compounds) are formed based on atomic structure (Bohr's model) and properties (valency)		
2	**C-1.2**	Investigates the nature and properties of chemical substances (distillation, crystallisation, chromatography, centrifugation, types and properties of mixtures, solutions, colloids, and suspensions)		
3	**C-1.3**	Describes and represents chemical interactions and changes using symbols and chemical equations (acid and base, metal, and non-metal, reversible, and irreversible)		
4	**C-2.1**	Applies Newton's laws to explain the effect of forces (change in state of motion: displacement and direction, velocity and acceleration, uniform circular motion, acceleration due to gravity) and analyses graphical and mathematical representations of motion in one dimension		
5	**C-2.2**	Explains the relationship between mass and weight using universal law of gravitation and connect it to laws of motion		
6	**C-2.3**	Manipulates the position of object and properties of lenses (focus, centre of curvature) to observe image characteristics and correspondence with a ray diagram, and extends this understanding to a combination of lenses (telescope, microscope)		
7	**C-2.4**	Manipulates and analyses different characteristics of the circuit (current, voltage, resistance) and mathematises their relationship (Ohm's law), and applies it to everyday usage (electricity bill, short circuit, safety measures)		
8	**C-2.5**	Defines work in scientific terms, and represents the relationship between potential and kinetic energy (conservation of energy) in mathematical expressions		
9	**C-2.6**	Demonstrates the principle of mechanical advantage by constructing simple machines (system of levers and pulleys)		
10	**C-2.7**	Describes the origin and properties of sound (wavelength, frequency, amplitude) and differences in what we hear as it propagates through different instruments		
11	**C-3.1**	Explains the role of cellular components (nucleus, mitochondria, endoplasmic reticulum, vacuoles, chloroplast, cell wall), including the semi-permeability of cell membrane in making cell the structural basis of living organisms and functional basis of life processes		

S. No.	Competency Code	Competency Details	Question Number	No. of items
12	**C-3.2**	Analyses similarities and differences in the life processes involved in nutrition (photosynthesis in plants; absorption of nutrients in fungi; digestion in animals), transport (transport of water in plants; circulation in animals), exchange of materials (respiration and excretion), and reproduction		
13	**C-3.3**	Describes mechanisms of heredity (in terms of DNA, genes, chromosomes) and variation (as changes in the sequence of DNA)		
14	**C-4.1**	Applies the knowledge of cellular diversity in organisms along with the ecological role organisms play (autotrophic or heterotrophic nutrition) to classify them into five-kingdoms		
15	**C-4.2**	Illustrates different levels of organisations of living organisms (from molecules to organisms)		
16	**C-4.3**	Analyses different levels of biological organisation from organisms to ecosystems and biomes along with interactions that take place at each level		
17	**C-4.4**	Analyses patterns of inheritance of traits in terms of Mendel's laws and its consequences at a population level (using models and/or simulations)		
18	**C-4.5**	Analyses evidences of biological evolution demonstrating the consequences of the process of natural selection in terms of changes: in allele frequency in population, structure, and function of organisms		
19	**C-5.1**	Explores how literature and the arts have influenced Science		
20	**C-5.2**	Examines a case study related to the use of Science in human life from the perspective of Social Sciences and ethics (e.g., Marie Curie, Jenner, treatment of patients with mental illness, the story of the atomic bomb, green revolution and GMOs, conservation of biodiversity)		
21	**C-5.3**	Applies scientific principles to explain phenomena in other subjects (sound pitch, octave, and amplitude in music; use of muscles in dance form and sports)		
22	**C-6.1**	Knows and explains the significant contributions of India to all matters (concepts, explanations, methods) that are studied within the curriculum in an integrated manner		
23	**C-7.1**	States concepts that represent the most current understanding of the matter being studied, ranging from mere familiarity to conceptual understanding of the matter as appropriate to the developmental stage of the students		
24	**C-7.2**	States questions related to matters in the curriculum for which current scientific understanding is well recognised to be inadequate		

S. No.	Competency Code	Competency Details	Question Number	No. of items
25	**C-8.1**	Develops accurate and appropriate models (including geometric, mathematical, graphical) to represent real-life events and phenomena using scientific principles and use these models to manipulate variables and predict results		
26	**C-8.2**	Designs and implements a plan for scientific inquiry (formulates hypotheses, makes predictions, identifies variables, accurately uses scientific instruments, represents data, primary and secondary, in multiple modes, draws inferences based on data and understanding of scientific concepts, theories, laws, and principles, communicates findings using scientific terminology)		

SUMMARY SHEET FOR VOCATIONAL EDUCATION

Summary: **Long Answer (LA)** **No.** _____ **Marks** _____ **Pattern of Options** **Grammar (One-word QS) and Writing Section**

Short Answer (SA) **No.** _____ **Marks** _____

Very Short Answer (VSA) **No.** _____ **Marks** _____ **Scheme of Sections** **Content Area**

Objective (O) **No.** _____ **Marks** _____

S. No.	Difficult Level	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Easy	25%	25%		
2	Medium	50%	25%		
3	Difficult	25%	50%		

S. No.	Abilities	Standard for Basic & Proficiency Level	Standard for Advanced Level	Question Number	No. of items
1	Awareness	50%	25%		
2	Sensitivity	25%	25%		
3	Creativity	25%	50%		

COMPETENCIES COVERED: VOCATIONAL EDUCATION

S. No.	Competency Code	Competency Details	Question Number	No. of items
1	C-1.1	Performs procedures competently through required tools/equipment.		
2	C-1.2	Differentiates between effective and non-effective practices in completing the task.		
3	C-2.1	Develops the following values while engaging in work: • Attention to detail • Persistence and focus • Curiosity and creativity • Empathy and sensitivity • Collaboration and teamwork • Willingness to do physical work		
4	C-3.1	Applies the acquired vocational skills and knowledge in a home setting.		

2.4 Writing Questions (Step 3)

Writing questions is the most critical step in developing a question paper. The quality of questions determines how well the paper assesses the targeted competencies and ensures balance. Each question must align with the test design and blueprint, adhering to the predetermined ability, content area, and difficulty level established in earlier stages.

As outlined in the blueprint, the required number of questions should be developed and recorded on item sheets. Each question, along with its specifications, is to be written on a separate sheet using a standardized template, as shown below:

Question Writing Sheet

Question Code:		Class:	
Subject:		Unit:	
Ability:		Marks:	
Mapped Competency:		Estimated Time:	
Type of Question:		Estimated Difficulty Level:	

The process of framing questions is so vital that the overall success of an examination depends on it. Furthermore, the quality of later stages, such as preparing the marking scheme, relies heavily on well-written questions.

While developing questions, item writers must consider several key aspects, including the characteristics of a good question and the strengths and limitations of different question types. Let's explore these aspects in detail.

2.4.1 CHARACTERISTICS OF EFFECTIVE AND FAIR QUESTIONS

Assessment questions are best thought of as communication between the assessor and the learner. Each assessment question should clearly, completely, and directly convey what knowledge the learner needs to deploy and what competencies, skills, or abilities the learner needs to demonstrate. Questions should provide all learners in the given test-taking population with a fair opportunity to communicate their competencies.

In keeping with the goal of clear, complete, and direct communication, all assessment items should display the following characteristics: CLARITY, UNITY, and ECONOMY.

CLARITY:

The prompt (or "stem") of an item should clearly and unambiguously point the learner to the desired competency. In other words, there should be only one way to correctly interpret the prompt. For instance, a question that features a picture of Rani Laxmi Bai and asks, "What is the significance of this picture?" is subject to too many possible interpretations to be a useful question.

Clarity also implies that some formulations of questions and question prompts may be consistently preferable to others. Questions that seek to elicit the same competency can (and in most cases should) have highly similar wording. Question writers should rely on proven formulations (sometimes called "task models") and should not introduce different wordings simply for the sake of interest or variety.

The language of the question should be specific and precise to clearly define the scope of the expected answer.

Additionally, the value points or marks assigned to a question must be clearly indicated. For questions with two or more parts, the marks allotted to each part should also be explicitly mentioned.

UNITY:

All parts of a question as a whole should aim only at the target competency. For instance, a mathematics item in the form of word problem should not include difficult vocabulary or unfamiliar situations, since these are unlikely to be part of a mathematics competency.

"All parts" must include the incorrect answer choices of a selected response question. One frequent mistake made by question writers is to create a prompt and stem that fits the desired competency but then add incorrect answer choices that can be eliminated by other means. For instance, a question intended to test a learner's knowledge of the causes contributing to a

significant historical event which has incorrect answers that can be eliminated because they refer to obviously fictional or concocted occurrences no longer measures the competency of causation.

Unity also implies that there should be a clear fit between the wording of the prompt and characteristics of the desired response. A science question that asks a student to "Evaluate the change in appearance of metallic iron when exposed to water in the presence of oxygen" is a poor fit with the desired response since there is no evaluation needed, only description.

ECONOMY:

Testing time is precious, so each question should only include as much material as is needed to point the learner toward the desired competency. Extraneous material such as decorative images or inessential information should be avoided. That said, it is sometimes necessary to employ more words than one might in a non-testing situation to make the task of the question clear. For instance, "Compare Anand's treatment of the independence movement to Tagore" would probably be understood in everyday speech, but it would be preferable, in a testing context, to explicitly spell out the comparison by asking instead "Compare Anand's treatment of the independence movement to Tagore's treatment of the independence movement."

Economy does not imply that all questions must be short. In order to test certain questions, it will often be necessary to supply a significant amount of material. However, all material supplied should be necessary for a student to display the desired competency.

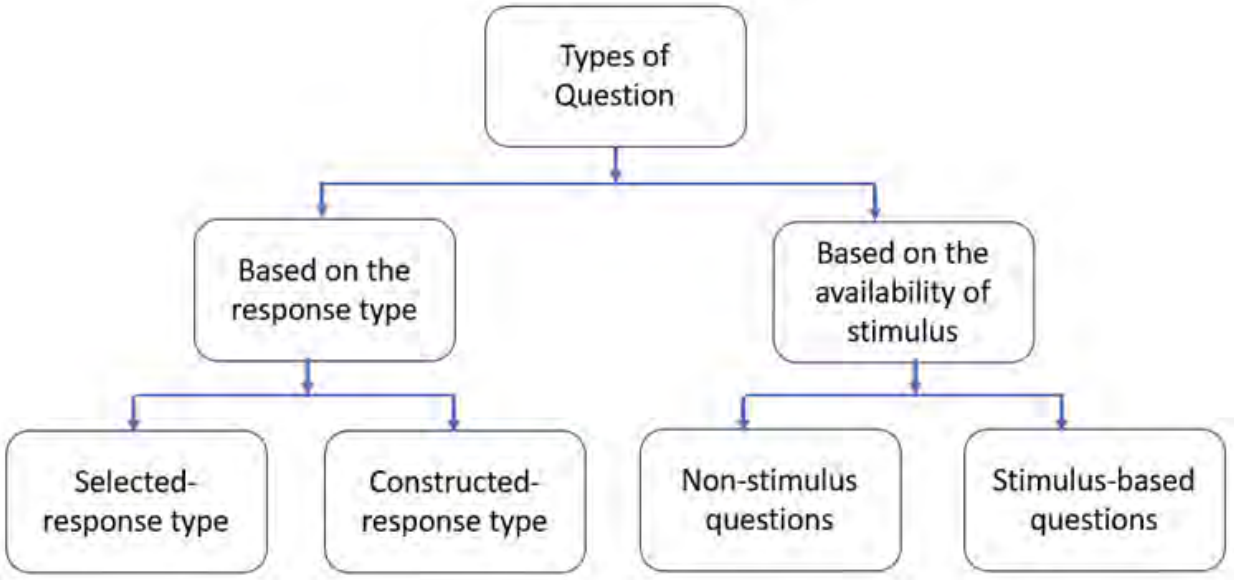
The principle of economy also implies that question writers should avoid trying to "teach through the test." Questions that gratuitously introduce new fundamental concepts or complex information beyond the needs of the target competency can compromise the integrity of the assessment.

2.4.2 STRENGTHS AND WEAKNESSES OF VARIOUS QUESTION TYPES

Question papers for school education board exams contain a variety of question types, ranging from traditional single-answer multiple choice questions to questions requiring the learner to construct an extended response. Different question types can appropriately serve different assessment aims, but the selection of question type or types for a Question paper should be a clear and conscious decision, based on the desired targets of measurement, student capabilities, and available scoring resources. This section of the manual discusses different types of questions, their strengths and weaknesses, and their recommended use cases.

Broadly speaking there are two ways to classify questions used on standardized assessments. The first is how the learner provides evidence of their competency, skill, and/or knowledge: either by **selecting** a response from a provided list of possible answers, or by **constructing** a response in reaction to a prompt.

The second mode of classifying questions pertains to whether the learner is expected to select or construct a response without any material beyond the prompt (a "non-stimulus question") or the learner is provided with additional material (a "stimulus-based question"). Stimulus material can take many forms: authentic texts, passages created explicitly for the test, images, graphics, charts, maps, data sets, etc. The choice of stimulus material (or the choice not to include it) should be guided by the target competency. For more guidance regarding stimulus material see section 3.5 below.



These two modes of classification are independent. In other words, there can be both stimulus-based and non-stimulus selected-response questions and both stimulus-based and non-stimulus constructed response questions. It is important to note that **all** four varieties of questions can be valid, depending on the measurement purposes they are intended to serve.

2.4.2.1 Selected response questions

Strengths or advantages of selected-response questions in general

Selected-response questions have certain advantages when compared to constructed-response questions. The first advantage is that the scoring is objective. In other words, there generally is (or should be) clear agreement as to the correct answer prior to the question being presented to learners. The second advantage of selected-response questions is that they can

almost always be scored and analysed more easily and require less time than constructed-response questions. The third is that they tend to be compact and take up a relatively small amount of testing time. Thus, a group of selected response questions can often be used to "cover" a larger number of competencies in the target domain.

Weaknesses or disadvantages of selected-response questions in general

Selected response questions have certain disadvantages when compared to -constructed-response questions. The first is the possibility of a learner guessing the correct answer given the available choices. The second is the difficulty of creating selected-response questions that address the highest ability or higher-order thinking skills. Although selected response questions can test more than factual recall, such questions must be carefully vetted and reviewed. The third disadvantage is related to the first two: selected-response questions, even well-crafted examples that truly measure higher-order thinking skills are sometimes dismissed or distrusted by educators, administrators, or the public, even though their use for measuring higher-order thinking skills has been extensively supported by research.

Despite these disadvantages, the use of selected-response questions as a significant part of standardized testing is almost always justified. When establishing the design and blueprint of a question Paper, teachers should strongly consider including a significant proportion of selected-response questions because of their greater coverage possibilities and their greater ease of scoring. Selected response questions come in three main types: single answer-multiple choice, multiple-answer multiple choice or alternate answer questions.

Single-answer multiple choice

This is the most familiar type of question, having been used successfully in a wide variety of subject areas since the middle of the twentieth century. As their name implies, each question of this type should have one and only one defensible correct answer to the prompt.

Recommended use cases: This question type can be used to test recall of discrete facts (though that will not typically be sufficient evidence of the target competency). It can also be used to test comprehension of material, inferences, application of concepts, mathematical operations, the ability to recognize valid explanations, as well as other competencies.

Multiple-answer multiple choice

As its name implies this question type is a multiple-choice question with more than one defensibly correct answer.

Recommended use cases: This question type is best employed to deal with complex concepts, multi-causal explanations, or competencies defined or taught as lists of desirable practices or behaviours.

In most cases, a single-answer multiple choice question can effectively target the same competency as a multiple-answer multiple choice, multiple-answer multiple-choice questions should only be used when necessary. Because this type of question is typically much less familiar to learners and is often mistaken for a traditional single-answer multiple-choice, questions of this type should clearly signal that multiple correct answers are possible. If practicable, these questions should be grouped together in a separate question paper section with a clear header that announces the question type. This type of question tends to function best when the prompt states how many of the answer choices are correct. At a minimum, the prompt should contain language that clearly indicates the need to look for multiple correct answers.

Alternate answer selected response

This question type is essentially a multiple-choice question with only two answer choices, one of which is correct. The most common form is the true-false question. True-false questions are relatively easy to construct and require less time and effort compared to other selected-response questions. True/False questions are effective for testing basic awareness and understanding of fundamental concepts.

Recommended use cases: Because of the high chance of guessing the correct answer, this question type should be used extremely sparingly, or not at all. In particular, it is not recommended to use true/false questions. It is possible, however, to construct valid questions that consist of two or more alternate choices. For instance, if the target competency was the learner's ability to use the correct sequence of tenses in formal written English, the following example of alternate-choice format could be valid:

"If I (have/had) seen your friend before I left the village, I (would/would have) given her your message."

Note that this has four possible permutations, only one of which is grammatically correct, thus it is functionally equivalent to a single-answer multiple choice question with four answer choices.

2.4.2.2 Constructed-Response Questions

Strengths or advantages of constructed-response questions in general

By requiring the test-taker to produce a correct response, rather than identifying one, constructed-response questions can eliminate the possibility of guessing. Constructed response questions can also more easily assess complex skills such as structured writing, scientific explanations or explications of mathematical or other reasoning. Short or very short constructed response questions can sometimes be written more quickly than selected response questions, since there is no need to craft wrong answer choices.

Weaknesses or disadvantages of constructed-response questions in general

The main disadvantage of constructed -response questions is that the need to score the learners' responses creates a degree of subjectivity. Unless the range and exact wording of possible credited responses is extremely limited, accurate scoring will depend on informed human judgment (either directly by a human scorer or built into a scoring program). Responding to constructed-response questions also typically takes more testing time, thus they usually provide smaller opportunities for content coverage. Apart from this, judging the answer and scoring is a time-consuming and often tiring process. Sometimes factors like handwriting, spelling mistakes and the teacher's impression of the student affects scoring. These types of questions, particularly essay-type ones, are less reusable than selected response questions. They are easier to remember and thus become insecure more quickly.

Importance of a rubric or marking scheme

Any constructed-response question used in a standardized test such as a school board question paper must have a rubric or scoring scheme created as part of the question-writing process. Rubrics may consist of a simple list of possible responses that would receive credit, or they may consist of descriptive statements specifying the characteristics of a successful response (For example: "The paragraph begins with a topic sentence that clearly and accurately states the main points made in the paragraph.").

Rubrics or marking schemes are primarily intended to provide scorers with clear and comprehensive guidance on what should and should not be credited in test-takers' responses. However, rubrics and marking schemes also have the important secondary purposes of signalling the intent of the question and serving as a check on the clarity of the question. Rubrics or marking schemes should be created with the first draft of the question and reviewed (and modified if necessary) alongside the question as part of subsequent reviews.

Types of Constructed Response Questions

The most common type of constructed-response question is according to the approximate length of the response the learner needs to produce to provide evidence of the target

competency. Based on the survey of existing state board QPs, constructed-response items can be usefully divided into the following categories:

Very short answer (VSA):

Responses to this type of question are typically expected to range from a single word or number to a short sentence or two. They sometimes take the form of "fill in the blank" questions. Learners are typically allotted 1-2 minutes of testing time for each question of this type.

Recommended use cases: Very short constructed-response questions are best employed to assess discrete pieces of knowledge, vocabulary, grammatical usage, simple mathematics problems, etc[4].

Short answer (SA):

Responses to this type of question are typically expected to range from 50-60 words. Learners are typically allotted 3 to 5 minutes of testing time for each question of this type.

Recommended use cases: Short constructed response question is best used to elicit multiple pieces of related information (e.g., significant factors contributing to a historical event), complex explanations or, comparisons or evaluations that cannot be adequately conveyed in a sentence or two. Short constructed-response questions can also be employed to elicit simple structured writing such as a sequence of events, a structured paragraph, or steps in a procedure.

Long Answer (E/LA):

Responses to this type of question are typically expected to have more than 100 words. Learners are typically allotted 8 to 10 minutes of testing time for each question of this type.

Recommended use cases: Long answer constructed response questions can be used for many of the same purposes as short answer CRs. The greater length of the expected response provides more opportunity to elicit structured writing.

[4] Please note regarding Very short answer-Constructed response (VSA-CR): Some VSA CR questions are constructed to have one and only one possible correct answer. In this case, it may be more cost-effective to reconfigure them as a selected-response question, since this will obviate the need for human scoring.

Value of coherence in long and short answer CRs:

These two types of questions should be structured to elicit responses that have some degree of coherence. A Long Answer CR that in effect asks learners to simply recount anything they may know or have learned about a topic is unlikely to result in fair or accurate measurement.

2.4.3 VALID USES AND DESIRABLE CHARACTERISTICS OF STIMULUS MATERIAL

The decision to include stimulus material should be guided solely by the demands of the target competency. For instance, if the question is intended to test map-reading skills, then a map is likely to be an appropriate stimulus. However, if the target competency is reading comprehension of text and a travel account is chosen as the text, including a map is likely to be superfluous and possibly distracting.

Stimulus material of any type should avoid extraneous elements. For instance, a picture of items with price tags in a shop used in a mathematics question does not need to include a picture of the shopkeeper. On the other hand, a question intended to target the competency of detecting significant patterns in complex data sets should include some data that does not fit the predominant pattern, since this is integral to the competency.

The length of textual stimulus should take into account available testing time and the number of questions based on the text. Having learners read a 300-word passage and answer only 2 multiple-choice questions based on it would represent a misallocation of testing time and test-taker effort. Text stimulus materials should also be at the appropriate grade level.

Visual stimulus material should be no more complex than necessary. Many learners have some degree of visual impairment and have difficulty processing visually dense images with lots of decorative features, small font sizes, or unconventional scripts. The ability to deal with such features is usually not part of the target competency.

In addition to choice of question type and decisions regarding stimulus material, another important consideration in question writing is that of Fairness and Accessibility

2.4.4 FAIRNESS AND ACCESSIBILITY IN QUESTION PAPER DESIGN

What is meant by “fairness” in the context of assessment? Fairness in testing can by defined by the validity of inferences drawn from test scores across different groups. Fairness is achieved when test scores are valid for all test takers regardless of their gender identity, socio-economic status, or other group affiliation not relevant to the abilities being tested.

A test or question paper can be considered "fair" to the extent that learners answer questions *correctly* because they possess the relevant knowledge and competencies AND the extent that learners answer *incorrectly* because they lack the relevant knowledge and competencies and not because of some other feature of the question. In other words, learners should get questions **right** for the **right reasons** and they should get them **wrong** for the **right reasons** as well! If learners are getting questions wrong for the wrong reasons, a test is said to display "construct-irrelevant variance," i.e. the test is measuring something other than what it was constructed to measure.

Fairness rules aim to prevent test items from causing three groups of barriers

2.4.4.1 Cognitive barriers (measuring irrelevant awareness)

Cognitive barriers can be introduced by unnecessary complication of language or the use of overly specialized terms. For instance, in a vocabulary test intended for a general population the agricultural term "plow" is probably acceptable, but "thresher" probably is not. Similarly, in the area of information technology "computer" would be an acceptable term for a general population, but "JPEG" is probably too specialized.

Cognitive barriers can also be introduced by assuming familiarity with knowledge or concepts that are not universally known. For instance, a mathematics item that assumed awareness of how a polo match is scored would risk being biased against learners of socio-economic groups not familiar with polo.

2.4.4.2 Affective barriers (eliciting strong emotions)

Determinations of fairness are directly connected to the purpose and audience of a test. A topic like food contamination might risk triggering an emotional response (an affective barrier) in a reading test for a general population but obviously would be relevant in a restaurant inspector's test. Affective barriers can be caused by the unnecessary introduction of upsetting or controversial topics. Such topics should be avoided unless they are absolutely necessary for valid measurement. These include **abortion**, **genocide**, **ethnic conflicts**, **gun control**, **child abuse**, **euthanasia**, etc.

Even in cases where the mention of such topics is warranted, they should be treated as neutrally as possible. For instance, on a history test, it might be appropriate to deal with past instances of ethnic or religious conflict, but graphic depictions of violence associated with these conflicts should still be avoided.

Some topics require careful presentation to avoid eliciting negative emotions, even if they are construct-relevant:

- **Advocacy**: Avoid taking sides unless needed for valid measurement.
- **Death, disease, and disasters**: Avoid graphic details unless necessary.

Gender Sensitivity

Question writers should ensure **gender neutrality** in test items, avoiding stereotypical language or assumptions (e.g., using "nurse" for both males and females, referring to women and men equally by name or role). Questions should not rely on the assumption that **gender** is universally regarded as binary; test questions should be inclusive of non-binary identities or at least avoid relying on strict binary definitions or linguistic constructions.

Example: "How many students are in the class?" should avoid assuming only male and female students.

Some examples of gender-biased terms to avoid and acceptable replacements:

Acceptable	*Unacceptable, Except in Historical or Literary Material*
Chair, leader	Chairman
Firefighter	Fireman
Human beings, people	Man, mankind
Mail carrier	Mailman
Sales representative	Salesman
Insurance agent	Insurance man
Supervisor	Foreman
Workers, personnel, labor	Manpower

Religion:

Religion has been identified as both a potential cognitive and affective source of construct-irrelevant variance. Test takers may have strong emotional attachments to religious views and could be upset by any perceived endorsement of opposing beliefs. Even subtle positive or negative references to religion—whether general or specific—can provoke strong reactions from certain groups. To ensure fairness, any necessary mention of religion in a test should be presented in an objective, factual, and neutral manner.

Special Considerations for Children

Tests for children (particularly younger children) should, if possible, avoid topics that deal with:

- **Disturbing or frightening content** (e.g., spiders, fire, abductions),
- **Inappropriate behavior** (e.g., cheating, fighting),
- **Family or moral issues** (e.g., divorce, family violence).

Sensitive topics that are construct-relevant for children or young children should be presented in the least upsetting way possible.

Following these principles will ensure that tests are fair, valid, and free from unnecessary emotional or cognitive bias, fostering equitable assessment for all test takers. The aim should be to give all learners the same chance to display the relevant knowledge and competencies.

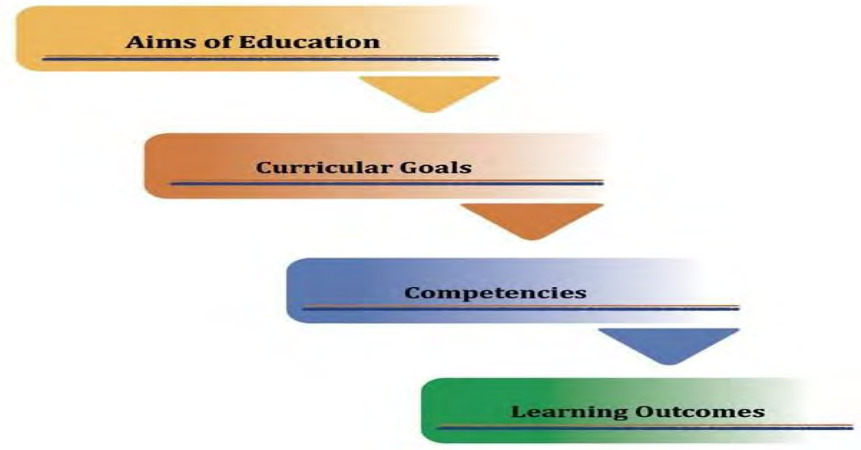
2.4.5 COMPETENCY BASED ASSESSMENT

Many assessment systems have focused on summative evaluations that test students' ability to recall information through rote memorization. While this approach provides a snapshot of a student's performance, it can fail to foster deep understanding, critical thinking, or the ability to apply knowledge in real-life contexts. The National Education Policy (NEP) 2020 emphasizes the urgency of transitioning from this outdated model to a more formative, competency-based approach.

According to NEP 2020, the new assessment framework must be regular, formative, and designed to promote learning and development. The focus is on higher-order skills such as analysis, critical thinking, and conceptual clarity. By prioritizing assessment for learning, this model ensures that teaching-learning processes are continually adapted to optimize outcomes for all students, fostering meaningful progress across educational stages.

NEP states that

> "The aim of assessment in the culture of our schooling system will shift from one that is summative and primarily tests rote memorization skills to one that is more regular and formative, is more competency based, promotes learning and development for our students, and tests higher order skills, such as analysis, critical thinking, and conceptual clarity. The primary purpose of the assessment will indeed be for learning; it will help the teacher and student, and the entire schooling system, continuously revise teaching learning processes to optimize learning and development for all students. This will be the underlying principle for assessment at all levels of education."
> [NEP 2020, 4.34]



2.4.5.1 Understanding Competency

The National Curriculum Framework (NCF) for School Education defines competencies as **measurable and observable learning achievements** that align with curricular goals. Competency based assessment evaluates not just what a student knows but how effectively they can apply their knowledge and skills in various contexts. This approach emphasizes ongoing development, encouraging students to build and refine their abilities progressively.

NCF School Education suggest:

> "Competencies are **learning achievements that are observable and can be assessed systematically**. These Competencies are derived from the Curricular Goals and are expected to be attained by the end of a Stage."
> NCF-FS 2022 and NCF-SE 2023

2.4.5.2 Why is Competency-Based Assessment Essential?

Competency based assessment aligns with the broader goals of 21st century education, which prioritize skills such as problem solving, adaptability, and creativity. Unlike rote memorization, which often results in shallow and temporary learning, this approach nurtures lifelong learners equipped to face complex challenges.

By focusing on meaningful learning outcomes, competency-based education transforms education into an empowering process, preparing students for real-world success. Through this shift, education systems can ensure that every learner achieves their potential and contributes meaningfully to society.

How learners are tested shapes how they will be taught. When educators know that learners will be faced by a summative exam consisting entirely of factual recall questions, they will emphasize rote memorization. On the other hand, true competency-based **assessment** will encourage competency-based **education**.

2.4.5.3 Assessment framing guidelines as applied to State Board Question Papers

S.no.	Quality check parameters	Mark (✓) if YES and Mark (✗) if NO
1	Assessment should measure achievement of Competencies and Learning Outcomes leading to attainment of Curricular Goals.	
2	Assessments should be Stage appropriate.	
3	Assessments should be constructive, developmental, and learning-focused.	
4	Assessments should focus on testing Higher Order Thinking Skills (HOTS).	
5	Assessments should promote reflective and self-regulated learning.	
6	Assessments should be transparent with clear criteria and expectations.	
7	Assessments should ensure inclusivity and equity in access and participation.	
8	Assessments should be designed to motivate and build confidence in learners.	
9	Assessments should accommodate student diversity	
10	Assessments should employ clarity and simplicity in item language and structure.	
11	Assessments should highlight the integration of technology where appropriate.	
12	Assessments should be aligned with contemporary societal and environmental issues.	
13	Assessments should encourage critical questioning and exploration of multiple perspectives	
14	Assessments should integrate interdisciplinary connections where applicable.	
15	Assessments should incorporate ethical and value-based considerations where applicable.	
16	Assessments should align with the principles of fairness and objectivity.	
17	Assessments should encourage creativity and innovation in responses where applicable.	
18	Assessments should allow for demonstration of skills through performance-based tasks.	
19	Assessments should emphasize problem solving and decision-making abilities.	
20	Assessments should promote ethical reasoning and cultural sensitivity.	
21	Assessments should include diverse question types to address multiple learning styles.	

2.4.5.4 Creating and reviewing competency-based questions

For an assessment to be considered competency-based, the individual questions must obviously be competency-based as well. In some subject areas, particularly mathematics, it can be relatively easy to create competency-based items, in part because there is virtually no distinction between content knowledge and procedural knowledge (or skill) in those fields. For instance, a competency statement that says, "Students can perform basic arithmetic (addition, subtraction, multiplication, division) with numbers up to three digits" immediately suggests how a question that tests that competency would be structured and what content it would have.

Most subjects, however, require more in-depth consideration and often choice of subject matter before a true competency-based question can be constructed. For instance, a history competency statement that says, "Students can understand the main causes of significant events in Indian history" requires a judgement as to what constitutes a "significant event." This judgment is likely to be informed by a combination of the content of widely used textbooks, the question writer's own experience as an educator, and any specific guidance provided by boards or other administrative bodies. Similarly, a language competency statement that says, "Students can comprehend important points from age-appropriate informative texts" requires several judgments.

Choice of question type

Once the competency has been identified with respect to subject matter and ability, the first choice of a question writer is what type of question can be used to measure this competency. In general, question writers should choose the shortest, simplest question-type commensurate with the target competency. For instance, if the target competency is knowledge of significant terminology in a technical field, measurement of this competency does not require a long-answer constructed-response type question. It could probably be measured adequately with single-answer multiple-choice or very-short-answer constructed-response questions. On the other hand, a test question aimed at the competency of creating and explaining an illustrative scenario for a particular concept or principle would almost certainly require a long-answer constructed-response type question and carefully specified scoring scheme or rubric.

Ability and question type

There is a widespread misconception that selected-response (multiple choice) questions are only suitable for measuring competencies at the most basic ability ("know"). This misconception has been reinforced by the long-standing practice of using multiple-choice question mostly to test factual recall or declarative knowledge. Research has shown, however,

that multiple-choice questions can in fact measure higher-order thinking skills. In practice, well-constructed multiple-choice questions have been successfully used to measure competencies at the abilities of awareness, sensitivity and creativity

The key term in the previous assertion is "well-constructed." For a multiple-choice question to provide valid evidence of higher-order thinking skills, the **entire question**, including the incorrect answer choices, must function together. For instance, a question intended to measure the ability to correctly apply literary terms to a fictional text passage should have incorrect answers that are all valid literary terms that the learner is likely to be familiar with. If, instead, the question writer invents imaginary terms or uses abstruse terms unlikely to have been seen by the test-taking population, then the question will no longer reliably measure the target competency, instead collapsing into a mere recognition task.

Given the greater economy and ease of scoring associated with multiple-choice items, they are often to be preferred to constructed-response items, especially if they can measure competencies at some of the higher level of abilities. Designers, blueprint makers, and questions writers should make every effort to include multiple-choice question, while also ensuring that they truly measure the competency.

Choice of context or setting

It is common practice to provide a context or setting for test questions. For instance, an arithmetic problem might ask the test-takers to imagine they are making purchases at a store, a reading comprehension question might have the test-takers read and extract information from a safety poster. Question writers typically provide a setting or context for one of the following reasons:

- The explicit requirements of the competency
- The desire to lessen test anxiety by using familiar settings
- The desire to improve test-taker engagement

The first reason is always valid. Whenever a competency calls out a particular setting or context (e.g. "Students can understand basic conversations on everyday topics."), then test questions targeting that competency should, as a matter of course, reflect that setting or context.

Attempting to alleviate test anxiety is a laudable goal and adding some material to place the test-taker in a familiar context may do no harm. However, any contextualizing material should be as brief as possible and should be checked carefully to make sure it is plausible, does not introduce unnecessary distraction, and is truly familiar. For instance, a question that targets test-takers' ability to compute area of rectangles might ask them to calculate the area

of a dining table. But if the dimensions of the table were implausibly large (e.g. 8 meters by 3 meters), this could work against the desire to create a familiar setting.

There is some research to support the idea that at least some test-takers are more engaged when test questions supply a concrete context. However, the same potential problems can arise as do when attempting to lessen anxiety by adding familiar contexts.

In general, writers of competency-based questions should be wary of gratuitously adding contexts or settings not required by the competency itself. Reviewers of competency-based questions should always ask themselves if any additional context is truly needed and whether it is likely to prove distracting to some students.

Stimulus-based or non-stimulus

When considering whether to use stimulus material, the question writer should be guided only by the explicit requirements of the competency and the principle of economy. Stimulus material should only be included in support of the target competency and only as much stimulus material as is needed. Writers should avoid "decorative" visual images in questions that can be solely verbal. Text passages or scenario descriptions should avoid providing large amounts of extraneous detail. Setting or scenario descriptions should provide the minimum necessary verbiage to put the test-takers in the scene.

The importance of question review

Even the most basic test questions intended for a large population should receive a thorough content and editorial and fairness review. This is particularly true of competency-based questions. Reviewers of such questions should ask themselves "Does the question as a whole, truly afford test-takers the opportunity to demonstrate the target competency by answering the question correctly?"; "Will test-takers who answer this question incorrectly truly lack the target competency?"; and "Are there features of the question that will interfere with its ability to measure the target competency?" If they can satisfy themselves on those three points, the question is likely to be part of a fair and valid assessment.

2.5 Preparation of a Question Paper (Step 4)

The preparation and assembly of a question paper is a crucial step in the assessment process, requiring careful planning, alignment with instructional goals, and attention to detail. A well-constructed question paper not only evaluates students' knowledge and skills effectively but also promotes a positive testing experience. This process begins with developing a comprehensive design blueprint that outlines the structure, content, and abilities to be assessed. The blueprint serves as a foundation for selecting and sequencing questions,

ensuring balanced coverage of topics, alignment with learning objectives, and adherence to prescribed difficulty levels. Additionally, the question paper must incorporate clear instructions and formatting that facilitate smooth administration and enable test-takers to focus on demonstrating their abilities without unnecessary confusion. A thoughtful approach to assembling a question paper ensures fairness, validity, and reliability while providing an opportunity for students to showcase their learning effectively.

There are several criteria for choosing the order in which questions appear in a question paper:

i. Type of questions
ii. Instructional objectives
iii. Content units,
iv. Difficulty level
v. According to the convenience of administration of the examinations

2.5.1 TYPE OF QUESTIONS

It is often desirable to group questions within a paper by type. Selected-response questions are frequently placed first since they are often easier and can serve as a confidence-builder to test takers.

2.5.2 INSTRUCTIONAL OBJECTIVES

It is sometimes desirable to place questions dealing with more basic competencies or skills (such as comprehension) earlier in the paper, so that test-takers encounter them in the same order in which they were instructed.

2.5.3 CONTENT UNITS

The order of questions within a topic or section should align with the logical flow of the content. If a specific sequence is commonly followed during teaching (e.g., chronological events in history), the same order should be maintained in the paper.

2.5.4 DIFFICULTY LEVEL

To the extent possible, it is desirable to present test-takers with easier questions before harder questions, assuming reasonably accurate estimates of difficulty can be made.

2.5.5 CONVENIENCE OF ADMINISTRATION

Administrative considerations can influence the order of questions. Separate sections for different question types (e.g., objective vs. subjective) reduce confusion. Grouping questions by response format minimizes the need for switching between answer sheets.

In addition to the questions themselves, question papers must have clear instructions and information for the test-takers. These directions should include:

- The suggested amount of testing time allotted for the paper as a whole or the section (if applicable)
- Means of indicating answers to questions (e.g. circling the correct answer choice, filling in a bubble on a separate answer sheet, writing an answer on blank lines, etc)
- The approximate length of expected answers to constructed-response questions and any specific required features of the response (e.g. "Write a structured paragraph with a topic sentence.")

If the question paper is divided into major topics or sub-topics (e.g. a science question paper that has different sections dealing with Biology, Chemistry and Physics), it can be helpful to the test-takers if the different sections are labelled as a kind of advance organiser.

By thoughtfully organizing questions and incorporating clear instructions, question papers become more accessible to test-takers, reduce ambiguity, and enhance the accuracy of assessment outcomes.

2.6 Preparing Scoring Key/Marking Scheme (Step 5)

Marking schemes are essential tools in educational assessment, serving as structured guides for teachers to evaluate constructed response questions systematically, fairly, and transparently. They enhance the consistency and objectivity of grading, ensuring that assessments reflect students' true understanding, skills, and effort. To develop robust and comprehensive marking schemes, the following principles should be observed:

2.6.1 ALIGNMENT WITH SPECIFIC ABILITY AND CONTENT

Marking schemes should directly correspond to the skills and content targeted by the item. Each item assesses specific learning objectives, whether conceptual understanding, analytical ability, problem-solving, or creativity. Marks should be distributed in a way that emphasizes these intended outcomes, ensuring that the evaluation reflects the focus of the question.

2.6.2 PROPORTIONAL ALLOCATION OF MARKS TO VALID VALUE POINT

The distribution of marks must be proportional to:

- **The scope of Expected Content**: Items requiring detailed explanations, multiple steps, or extensive reasoning should carry more weight.

- **The Time Investment Required**: Marks should also align with the estimated time students need to respond. For example, higher marks should be allocated to items involving multi-step problem-solving or complex analysis.

This approach ensures fairness, as students are appropriately rewarded for effort and depth in their responses.

2.6.3 RECOGNITION OF VALID AND EXTENDED ANSWERS

While the marking scheme should outline correct and valid answers, it should also allow flexibility to reward responses that:

- Go beyond the scope of the syllabus.
- Demonstrate creative thinking, critical analysis, or unique perspectives.

Rubrics should include provisions for alternative valid responses that are not explicitly mentioned but reflect accurate reasoning or advanced understanding. This flexibility encourages innovation and rewards students who think critically or apply knowledge in novel ways.

2.6.4 CLEAR GUIDELINES TO MINIMIZE SUBJECTIVITY

Subjectivity in evaluation can compromise fairness. To eliminate this, marking schemes should provide:

- **Detailed Instructions**: Clear descriptions of what constitutes a correct or complete answer.
- **Explicit Criteria**: A breakdown of marks for different components of an answer (e.g., structure, accuracy, originality).
- **Analytic rather than holistic scoring**: To the extent possible, rubrics should use a series of binary determinations (e.g. Valid thesis statement present or not? Presented as paragraphs organized around topic sentences or not?) rather than relying on sweeping statements about the overall quality of a response (e.g. "Response is a highly persuasive essay.")
- **Exemplar Responses**: Model answers, or sample responses can help evaluators understand the expected standard and reduce ambiguity.

Consistency in grading is achieved when evaluators have a precise reference point, minimizing the impact of personal biases or interpretations.

2.6.5 DETAILED VALUE POINTS

Each marking scheme should specify key value points for every question. These value points:

- Highlight the critical elements of an answer that must be included to earn marks.
- Provide a clear framework for evaluators to assess answers systematically. For instance, a long-answer question in science might include value points for defining terms, explaining principles, and providing relevant examples or diagrams.

By breaking responses into measurable components, value points ensure that the marking process is both systematic and exhaustive.

2.6.6 AVOIDANCE OF NEGATIVE MARKING IN RUBRICS

Marking schemes should focus on rewarding correct or partially correct responses, rather than penalizing errors. Defining negative marks within the rubric can lead to confusion and may discourage students from attempting questions. Instead:

- Errors should be noted in feedback without disproportionately impacting overall marks.
- Rubrics should emphasize use a positive approach that highlights areas for improvement while acknowledging effort and valid attempts.
- Rubrics should distinguish and to the extent possible specify the difference between serious conceptual or procedural errors that rightly prevent a response from receiving full credit and minor errors that do not detract from the overall quality of the response.
- It is good practice to phrase the language of the rubric so that it focuses on the characteristics of the RESPONSE rather than on the abilities or shortcomings of the STUDENT.

2.6.7 SUBJECT-SPECIFIC CONSIDERATIONS

Different subjects demand distinct response formats, reflecting the unique nature of their content and objectives:

- **Mathematics and Sciences**: Responses should be evaluated for accuracy, logical reasoning, and stepwise problem-solving. Diagrams and units, where applicable, must be considered.
- **Languages**: Focus on content, structure, grammar, and creativity. Subjective answers (e.g., essays) require well-defined rubrics with weightage for different aspects like coherence and originality.
- **Humanities**: Assessments should consider argumentation, critical analysis, and use of evidence. Marking schemes should emphasize clarity and depth of reasoning.

Custom templates for each subject should be prepared, incorporating examples and response structures that reflect subject-specific demands.

2.6.8 ENCOURAGING FAIR AND INCLUSIVE ASSESSMENT

Marking schemes should be inclusive and designed to fairly assess the diverse abilities and strengths of students. Some strategies to ensure this include:

- Using language in the rubrics that is clear and accessible.
- Accommodating alternative valid approaches to problem-solving or reasoning.
- Providing additional guidance for evaluators on how to handle unique or unexpected responses.

A well-constructed marking scheme serves as an effective assessment, ensuring that grading is consistent, objective, and reflective of students' efforts and abilities. By adhering to these principles—emphasizing alignment with learning objectives, proportionality, flexibility, and subject-specific nuances—teachers can foster a more transparent and equitable evaluation process. Comprehensive marking schemes not only improve the reliability of assessments but also enhance their role as tools for learning and growth.

2.6.9 CHECKLIST FOR DEVELOPMENT OF MARKING SCHEME

S.No.	Quality check parameters	Mark ✓ if Yes and Mark ✕ if No
1	Are the marks distributed according to the specific skills assessed (e.g., analysis, application, creativity)?	
2	Does the marking scheme explicitly state the learning objectives each question addresses?	
3	Are the criteria aligned with the syllabus and lesson plan?	
4	Is there a distinction between core knowledge and higher order thinking skills in the distribution of marks?	
5	Are instructions for skill-based evaluation (e.g., critical thinking, problem-solving) explicitly defined?	
6	Is the weightage of each question proportionate to its relevance in the curriculum?	
7	Are interdisciplinary skills (e.g., communication in science or logic in humanities) incorporated where relevant?	
8	Are higher marks allocated to questions requiring detailed reasoning, multi-step problem-solving, or analysis?	
9	Is the time required to answer each question considered when assigning marks?	
10	Are questions with similar difficulty levels assigned consistent marks across the paper?	
11	Is the proportionality between the marks and effort required for subjective questions (e.g., essays) reasonable?	
12	Are correct answers, as defined in the marking scheme, included explicitly?	
13	Does the marking scheme account for alternative approaches or methods to solve a problem?	
14	Are evaluators instructed to reward evidence-based reasoning, even if the final answer is incorrect?	
15	Are criteria provided for partial credit in multi-step problems or partially correct responses?	
16	Are extended, well-reasoned answers given appropriate recognition, even if unconventional?	
17	Are clear and detailed marking instructions provided for each question?	
18	Are criteria for awarding marks specific enough to minimize evaluator interpretation?	
19	Are common misconceptions or errors outlined with guidance on how to address them?	
20	Does the scheme explicitly describe the level of detail required in student responses?	

21	Are descriptors provided for awarding full, partial, or no credit in subjective questions?	
22	Are key value points identified for every part of the question?	
23	Is the weightage of each value point clearly defined in the marking scheme?	
24	Are marks allocated for supporting elements (e.g., relevant examples, proper formatting, or diagrams)?	
25	Are correct procedures or methods (e.g., showing work in math) part of the evaluation criteria?	
26	Is the importance of clarity, accuracy, and organization addressed in the value points?	
27	Are templates tailored to the nature of the subject (e.g., numerical accuracy for math, structure for essays)?	
28	Are unique evaluation criteria provided for different types of questions (e.g., objective vs. subjective)?	
29	Are discipline-specific skills (e.g., grammar in languages, experiment design in sciences) explicitly addressed?	
30	Are technical or format requirements (e.g., labelled diagrams in biology) included in the rubric?	
31	Are rubrics differentiated for creative or exploratory disciplines (e.g., art, literature)?	
32	Is the balance between content, structure, and originality defined for subjects like humanities or languages?	
33	Are subject-specific nuances (e.g., stepwise logic in math, evidence-based reasoning in history) emphasized?	
34	Are marking schemes reviewed for inclusivity and sensitivity to diverse evaluator backgrounds?	

2.7 Question paper Analysis (Step 6)

After drafting a question paper, it is important to ensure that it aligns with the blueprint and design. To determine the degree of alignment, the paper reviewer must analyse each question against key criteria, such as the cognitive skills it assesses, the specific skills it measures, the unit it belongs to, the question type, the expected answer length, the average time required by a student to answer, and its estimated difficulty level. The following checklist can be useful in conducting this review:

1. Does the question paper carry weightages for all the abilitys same as given in the design?
2. Does the question paper carry weightages for all content areas same as given in the design?
3. Does the question paper carry weightages for all forms of questions same as given in the design?
4. Does the question paper carry weightages for different difficulty levels same as given in the design?
5. Does the question paper follow scheme of sections same as given in the design?
6. Does the question paper carry pattern of options same as given in the design?
7. Does the question paper have allotted time to attempt the paper same as given in the design?
8. Is the content tested the same as indicated in blueprint?
9. Is the unit tested the same as indicated in blueprint?
10. Does the question really test the intended abilitys?
11. Does the question measure the intended competency?
12. Is the question set in the same form as shown in blueprint?
13. Does the wording of question evoke the intended response?
14. Does the language of question clarify the scope of answer?
15. Does the question carry the same marks indicated in the blueprint?

Additionally, the proforma given below can also be used for a systematic cross-check. This proforma is divided into two parts:

Part I: Collects general information about the question paper, such as the name of the board, examination details (name and year), class, subject, question paper number, maximum marks, duration, and any sections present.

Part II (Question-wise Analysis): Focuses on reviewing each question to ensure it aligns with the test design and blueprint. This includes an evaluation of the ability being tested, the content area, question format, marks allocated, estimated time for completion, and difficulty level. Comments or remarks are also recorded for each question, categorizing them as "OK," "Good," "Needs Improvement," or "Reject," based on the analysis.

2.7.1 PART I: GENERAL INFORMATION

1. Name of the Board:
2. Name of Examination:
3. Year of Examination:
4. Class:
5. Subject:
6. Question Paper Number:
7. Maximum Marks:
8. Time Allotted:
9. Sections, if any:

2.7.2 PART II: GENERAL ITEM ANALYSIS OF A QUESTION PAPER

Ques. No.	Objective	Content	Unit/Sub-Unit No.	Form of Question	Marks Allotted	Estimated time (min)	Estimated Difficulty Level
1							
2							
3							
4							
5							
6							
7							
8							

[1.]Performa for question paper Analysis

This analysis helps identify any imbalances in the question paper, enabling necessary modifications to improve its quality.

2.7.3 QUALITATIVE ANALYSIS OF A QUESTION PAPER

A question paper's effectiveness largely depends on the quality of its individual items. The qualitative analysis of these items focuses on various parameters, including language, structure, content, question format, options, allocated time, and marking scheme. Each parameter must be critically evaluated to ensure the following:

- **Language of the Question**: Is it simple, clear, precise, unambiguous, and free from redundancy? Can it be easily translated and comprehended by students?
- **Structure of the Question**: Does it include directional words that clearly define the scope and style of the expected answer? Is the stimulus provided functional and acceptable?
- **Content of the Question**: Is it accurate, relevant to the syllabus, and focused on significant learning points rather than trivial details? Does it include adequate value points or criteria for awarding marks?
- **Form of the Question**: Is it appropriate for the purpose (e.g., selection or construction), marks allocation, and the time required? Does it adequately sample the content?
- **Comparability of Options**: Are the options comparable in terms of objectives, content, difficulty level, time required, and marks allocation?
- **Time Allocation**: Is the time allocated sufficient for answering each question?
- **Marking Scheme:** Is the distribution of marks appropriate and in line with the value points/criteria? Does the marking scheme align with the answers provided by examinees?

Any defects or areas for improvement identified during the analysis should be documented in the table below:

Question no.	Point of Analysis	Detailed Observation/Remark (if any)

These observations guide the refinement process to ensure the question paper is balanced, fair, and of high quality. For reference, a supportive checklist is included to assist in systematically reviewing these parameters.

2.7.4 QUESTION-WISE REVIEW CHECKLIST

S.No.	Quality check parameters	Mark ✓ if yes and mark X if No
1	Item Structure	
1.1	Is the stimulus clear, concise, and directly relevant to the learning objective?	
1.2	Does the stimulus provide sufficient context and information to enable the test-taker to respond effectively?	
1.3	Does the stimulus avoid unnecessary jargon, overly complex language, or irrelevant information?	
1.4	If visuals (e.g., graphs, charts, or images) are used, are they clear, labelled, and relevant to the item?	
1.5	Does the stem clearly communicate what is being asked, without ambiguity?	
1.6	Is the stem phrased as a complete question or task that aligns with the stimulus?	
1.7	Does the stem avoid leading the test-taker to the correct answer through unintended cues?	
1.8	Is the language in the stem accessible and appropriate for the target audience's skill level?	
1.9	Are the distractors plausible and free from obvious errors or impossibilities?	
1.10	Do the distractors reflect common misconceptions or errors that align with the learning objective?	
1.11	Are all options (distractors and the key) written in parallel structure, with similar length and complexity?	
1.12	Do the distractors avoid overlapping content with the key or other options?	
1.13	Is there no use of ambiguous options such as "all of the above" or "none of the above"?	
1.14	Is the key indisputably correct, with no ambiguity?	
1.15	Does the key align precisely with the stimulus and stem?	
1.16	Is the key not overly detailed, technical, or distinct in tone compared to the distractors?	
1.17	Does the key address the specific skill or knowledge being assessed?	
1.18	Is the item clear and free from ambiguity, ensuring a single, consistent interpretation?	
1.19	Does the item encourage detailed, thoughtful, and structured responses?	

1.20	Are expectations for responses clearly communicated (e.g., "Provide at least three reasons," "Analyze the situation using the given framework")?	
1.21	Does the item align with the ability being assessed (e.g., application, analysis, evaluation?	
1.22	Does the case provide relevant, real-world context that aligns with the learning objectives?	
1.23	Is the case information structured logically, with clear and concise details?	
1.24	Are there any extraneous or irrelevant details in the case that might confuse test-takers?	
1.25	Does the case require the application of knowledge to a new or complex situation?	
1.26	Are all key terms and references within the case clearly defined?	
1.27	Are prompts sequenced logically if multiple parts are involved (e.g., "Describe the problem, propose a solution, evaluate its feasibility")?	
1.28	Are response formats specified where applicable (e.g., "Write a short essay," "List three recommendations")?	
1.29	Are scaffolding questions provided if the item is designed to assess complex skills?	
1.30	Do any hints or guiding questions support but do not directly answer the main question?	
2	Content alignment	
2.1	Is the item aligned with the specified ability?	
2.2	Is the item aligned with the specified instructional objective?	
2.3	Is the content consistent with subject-specific benchmarks?	
2.4	Does each objective have adequate and relevant content supporting it?	
2.5	Does each item have a clear instructional purpose that aligns with assessment objectives?	
3	Content Validity	
3.1	Is the item clear, and factually and conceptually, correct?	
3.2	Does the item reflect current and accurate subject-specific conventions and standards?	
3.3	Does the item avoid cultural, gender, or socioeconomic stereotypes?	
3.4	Is the numerical data in the item realistic and error-free?	

3.5	Are any references or citations used in the item accurate and relevant?	
3.6	Is the content relevant to the learner's context, age, and prior knowledge?	
4	Language	
4.1	Is the vocabulary used in the item contextual, age-appropriate, and sensitive to bias?	
4.2	Is the language of the item concise and free from unnecessary complexity?	
4.3	Is technical terminology, if used, explained or contextualized?	
4.4	Does the item minimize ambiguity to avoid misinterpretation?	
5	Quality Concerns in Item Art (Graphics) and Other Stimuli	
5.1	Is art/stimulus necessary, relevant, and useful to answer the item?	
5.2	Is art/stimulus likely to be interesting and engaging to students?	
5.3	Is art/stimulus pitched at an appropriate grade, age, or reading level?	
5.4	Is the art/stimulus clear, accurate, and sufficient to answer the item?	
5.5	Does the art/stimulus avoid being confusing or overwhelming?	
5.6	Does the art/stimulus contain appropriate and accurate labels?	
5.7	Is the art/stimulus free of errors or inconsistencies that might mislead students?	
5.8	Does the art/stimulus avoid clueing the correct answer to an item?	
5.9	Is the art/stimulus significantly free from copyright or intellectual property issues?	
6	Item Bias and Sensitivity Issues	
6.1	Is the item accessible to the greatest number of test-takers, including those with diverse abilities and backgrounds?	
6.2	Is the item free from bias in the areas of gender, caste, religion, socio-economic class/status, regional diversity, age, culture, or physical appearance?	
6.3	Is the item language unbiased towards a particular linguistic group?	
6.4	Is the item sensitive to the needs of special-needs groups, including physical, cognitive, or sensory disabilities?	

6.5	Does the item avoid offensive, disturbing, or controversial information that may alienate or distress test-takers?	
7	Formatting	
7.1	Is the font size appropriate (e.g., 12-14 pt for text, larger for headings) and consistent throughout?	
7.2	Is a clear and legible font type used (e.g., Arial, Times New Roman, Calibri)?	
7.3	Are items spaced adequately to avoid visual clutter (e.g., 1.15–1.5 line spacing)?	
7.4	Is there sufficient white space around each item to improve readability?	
7.5	Are margins uniform and appropriately sized?	
7.6	Are the stimulus, stem, and options distinctly separated and visually organized?	
7.7	Are all items numbered consistently and sequentially (e.g., Arabic numerals or letters for sub-parts)?	
7.8	For multi-part questions, are sub-questions appropriately labelled (e.g., a, b, c)?	

2.8 Moderation of Question Paper (Step 7)

Moderation is the process of reviewing and refining question papers to ensure they are high-quality, relevant, and practical for exams, whether at the school level or for board examinations. It involves examining the overall structure of the paper to confirm that it aligns with the exam's blueprint and design and meets the required standards.

This step is crucial because errors or biases can sometimes find their way into the paper-setting process, often due to personal preferences or a lack of technical knowledge. Moderation helps identify and correct such issues, making the question paper a reliable and valid tool for assessment.

Effective moderation requires subject experts who have a thorough understanding of paper-setting techniques and the exam's instructional objectives. These experts ensure the question paper serves its purpose, is practical to administer, and can be accurately scored and processed.

Key elements reviewed during moderation include:

1. The questions.
2. The answer key and marking scheme.
3. Instructions provided to students.
4. The paper's format and structure.
5. The clarity and language of questions.
6. Adherence to the specification as laid out in the Design and Blueprint

For school-based exams like term-end or summative assessments, moderation is often done internally by subject teachers and senior educators. However, for board exams, which have higher stakes, external moderators are necessary to maintain confidentiality and ensure unbiased reviews. External moderators must:

- Be different from the paper setter.
- Have deep subject knowledge.
- Be familiar with the syllabus and instructional goals.
- Be trained in paper-setting processes.
- Have proven skills in reviewing exam items.

Moderators should not make arbitrary changes or superficial edits. Moderators should respect the work of the paper setter and only make necessary adjustments to correct typographical errors and repair "fatal flaws" that would prevent the questions from functioning as intended. Untrained moderators who lack expertise in instructional objectives and paper-setting principles can unintentionally compromise the quality of a well-designed question paper.

Moderators should have access to the Exam Design and Blueprint so they can accurately gauge the contents of the questions against the needs of the design. They should also be provided with information about the classification of individual questions with respect to subject areas, specific competencies, and abilities. Classification information can be provided either in the form of a text document or a spreadsheet but should clearly identify the intended classification for each individual question in the question paper.

Moderators should also be given sufficient time to do a thorough review of both the questions and the marking schemes. If the moderator wishes to propose extensive changes, it may be necessary to have a "resolution meeting" during which possible changes are discussed and agreed upon.

Moderation is a vital step in the process of creating high-quality question papers and should never be an afterthought.

3 Credit Distribution at the Secondary Stage

The credit structure for the Secondary Stage (Grades 9 to 12) has been aligned with the National Credit Framework (NCrF), ensuring that students' academic and applied learning is acknowledged comprehensively. Each grade features a carefully balanced distribution of credits across subject-specific learning, MOOCs, and project-based learning. Credit points are calculated by multiplying the assigned credits with the NCrF level for the respective grade.

Grades 9 and 10

In Grades 9 and 10, students earn a total of 40 credits per grade. Out of these, 32 credits are earmarked for subject-specific learning, with an explicit inclusion of vocational skilling within both formative and summative assessments.

- **Grade 9**

  70% of subject-specific credits are allotted to formative assessment (21.9 credits + 0.5 vocational skilling) and 30% to summative assessment (9.1 credits + 0.5 vocational skilling), placing strong emphasis on continuous and classroom-based feedback. In addition to subject-specific learning, Grade 9 includes a census-based assessment (2 credits), MOOCs (2 credits), and projects/research work (4 credits, including 1 credit for vocational work). At NCrF Level 2.5, Grade 9 students can accumulate up to 100 credit points.

<table>
<tr><th colspan="6">Grade 9</th></tr>
<tr><th colspan="2">Criteria</th><th colspan="3">Credits</th><th>NCrF Level</th><th>Credit Points</th></tr>
<tr><td rowspan="4">Subject-specific Learning</td><td rowspan="4">1080 Hours of Learning</td><td>Formative Assessment</td><td colspan="2">22.4 (70% of 32 credits) = 21.9 (rest) + 0.5 (Vocational Skilling)</td><td>2.5</td><td>56</td></tr>
<tr><td rowspan="2">Summative Assessment</td><td>Subject-specific</td><td>9.6 (30% of 32 Credits) = 9.1(rest) + 0.5(Vocational Skilling)</td><td>2.5</td><td>24</td></tr>
<tr><td>Census-based Assessment</td><td>2 Credits</td><td>2.5</td><td>5</td></tr>
<tr><td>MOOCs</td><td colspan="3">2 Credits (1 credit = 15 hours)</td><td>2.5</td><td>5</td></tr>
<tr><td colspan="2">Projects & Research</td><td colspan="3">4 Credits (120 hours) (1 Credit for Vocational)</td><td>2.5</td><td>10</td></tr>
<tr><td colspan="2">Total Credits</td><td colspan="3">40</td><td>Total Credit Points</td><td>100</td></tr>
</table>

- **Grade 10**

  The division between formative and summative assessment is equal, with each carrying 15.5 credits + 0.5 vocational skilling (50:50 ratio). Beyond subject-specific credits, students complete MOOCs (4 credits) and projects/research (4 credits, including 1 credit for vocational work). At NCrF Level 3, Grade 10 students can accumulate up to 120 credit points.

Grade 10					
Criteria		**Credits**		**NCrF Level**	**Credit Points**
Subject-specific Learning	1080 Hours of Learning	Formative Assessment	16 (50% of 32 Credits) = 15.5 (rest) + 0.5 (Vocational Skilling)	3	48
		Summative Assessment	16 (50% of 32 Credits) = 15.5 (rest) + 0.5 (Vocational Skilling)	3	48
MOOCs		4 Credits		3	12
Projects & Research		4 Credit (1 Credit for Vocational)		3	12
Total Credits		**40**		**Total Credit Points**	**120**

Grades 11 and 12

With increased academic depth, both Grades 11 and 12 carry a total of 44 credits each. As per the National Credit Framework (NCrF), students are expected to earn a minimum of 40 credits per academic year.

- **Grade 11**

  Of the 36 subject-specific credits, 40% (13.9 credits + 0.5 vocational skilling) are assigned to formative assessment, while 60% (21.1 credits + 0.5 vocational skilling) go to summative assessment. In addition, students complete MOOCs (4 credits) and projects/research (4 credits, including 2 credits for vocational work). At NCrF Level 3.5, Grade 11 students can earn up to 154 credit points.

Grade 11

Criteria		Credits		NCrF Level	Credit Points
Subject-specific Learning	1080 Hours of Learning	Formative Assessment	14.4 (40% of 36 Credits) = 13.9 (rest) + 0.5 (Vocational Skilling)	3.5	50.4
		Summative Assessment	21.6 (60% of 36 Credits) = 21.1 (rest) + 0.5 (Vocational Skilling)	3.5	75.6
MOOCs		4 Credits		3.5	14
Projects & Research		4 Credits (2 Credits for Vocational)		3.5	14
Total Credits		**44**		**Total Credit Points**	**154**

- **Grade 12**

The subject-specific structure remains at 36 credits, but the weightage shifts to 30% (10.8 credits) for formative and 70% (25.2 credits) for summative assessment. MOOCs and projects/research contribute 4 credits each, bringing the total to 44 credits. At NCrF Level 4, Grade 12 students can accumulate up to 176 credit points.

Grade 12					
Criteria		**Credits**		**NCrF Level**	**Credit Points**
Subject-specific Learning	1080 Hours of Learning	Formative Assessment	10.8 (30% of 36 Credits) = 10.3 (rest) + 0.5 (Vocational Skilling)	4	43.2
		Summative Assessment	25.2 (70% of 36 Credits) = 24. 7 (rest) + 0.5 (Vocational Skilling)	4	100.8
MOOCs		4 Credits		4	16
Projects & Research		4 Credits (2 Credits for Vocational)		4	16
Total Credits		**44**		**Total Credit Points**	**176**

4 Conclusion

As this document indicates, the task of creating balanced, high-quality question papers is a complex and multifaceted endeavor. It involves a systematic process that requires the coordinated efforts of numerous individuals, each playing distinct and crucial roles. The creation of assessments should never be an afterthought or conducted in a hurried manner. Instead, it demands careful planning, meticulous attention to detail, and a commitment to educational integrity.

The process begins with the clear definition of learning objectives and the development of a comprehensive blueprint that outlines the scope and structure of the assessment. This blueprint serves as a guide for item writers, who craft questions that are aligned with the specified objectives and designed to measure a range of cognitive skills, from basic recall to higher-order thinking.

Once the questions are developed, they undergo rigorous review to ensure clarity, fairness, and alignment with the intended learning outcomes. This step is critical to identify and eliminate any potential biases or barriers that could disadvantage certain groups of students. Pretesting, when possible, also provides valuable data on the performance of individual items, which can be used to refine and improve the assessment.

The final stages of the process include assembling the question paper, preparing the scoring key or marking scheme, and ensuring that all logistical aspects of test administration are in place. Throughout this process, it is important to maintain a focus on the ultimate goal: creating assessments that provide all learners with a fair opportunity to demonstrate their competencies.

In conclusion, the development of high-quality assessments is a deliberate and collaborative effort that requires expertise, precision, and a deep understanding of educational principles. While decisions about specific aspects of the process and the final product may vary, the overarching objective remains the same: to create assessments that are valid, reliable, and equitable, thereby supporting the goal of achieving comparable learning outcomes for all students.